Supplementary Information: Posterior Information Distortion, Effective Sample Size, Corrected Covariance, and Bias

1. Log-Posterior and Score Definitions

Consider a trajectory of T sequential measurement intervals with data $D = (y_1, \dots, y_T)$, and a Gaussian prior $\pi(\theta)$ on the parameter vector $\theta \in \mathbb{R}^d$. Throughout this supplement, all derivatives, scores, and Hessians are taken in the transformed coordinates in which π is Gaussian; no Jacobian corrections appear. We restrict to Gaussian priors with fixed (non-random) hyperparameters, so the prior log-density and its derivatives are deterministic functions of θ only. As in `supplement_information_distortion_main.tex`, all expectations and covariances are taken under the frequentist sampling distribution of D at a fixed true θ (the role of the prior here is to regularize curvature and shift the score mean, not to randomize θ).

Notation bridge to the Likelihood supplement. We adopt the subscripts lik, prior, post to distinguish the data, prior, and combined contributions. The likelihood-side objects are renamings of the unsubscripted objects of the Likelihood supplement:

$$\mathbf{H}_{\text{lik}} \equiv \mathbf{H}, \qquad \mathbf{JT}_{\text{lik}} \equiv \mathbf{J}_{\text{total}}, \qquad \mathbf{JS}_{\text{lik}} \equiv \mathbf{J}_{\text{sample}}, \qquad S_{\text{lik}} \equiv S, \tag{1}$$

all of Sections 1–3 of `supplement_information_distortion_main.tex`. We also write $\ell(\theta) = \log L(\theta)$ as in the Likelihood supplement.

Log-posterior and posterior score. The log-posterior decomposes as

$$\log p(\theta \mid D) = \ell(\theta) + \log \pi(\theta) + \text{const}, \tag{2}$$

where $\ell(\theta) = \sum_{t=1}^{T} \ell_t(\theta)$ with $\ell_t(\theta) = \log p(y_t \mid \theta, y_{1:t-1})$, and the additive constant $-\log p(D)$ does not depend on θ. The posterior score is

$$S_{\text{post}}(\theta) = \nabla_\theta \log p(\theta \mid D) = S_{\text{lik}}(\theta) + s_\pi(\theta), \tag{3}$$

where

$$S_{\text{lik}}(\theta) = \sum_{t=1}^{T} s_t(\theta), \qquad s_t(\theta) := \nabla_\theta \ell_t(\theta), \qquad s_\pi(\theta) := \nabla_\theta \log \pi(\theta). \tag{4}$$

For the Gaussian prior $\pi(\theta) = \mathcal{N}(\mu_{\text{prior}}, \Sigma_{\text{prior}})$ with precision $\mathbf{H}_{\text{prior}} = \Sigma_{\text{prior}}^{-1}$, this gives the explicit form

$$s_\pi(\theta) = -\mathbf{H}_{\text{prior}}(\theta - \mu_{\text{prior}}). \tag{5}$$

Per-interval vs. global decomposition. The likelihood score retains the per-interval structure inherited from the trajectory model, while the prior contributes a single global term that is not indexed by measurement interval. We write

$$S_{\text{post}}(\theta) = \Big(\sum_{t=1}^{T} s_t(\theta)\Big) + s_\pi(\theta), \tag{6}$$

distinguishing the per-interval sum from the global prior term.

Prior as a global deterministic observation. Throughout this supplement we treat the prior $\pi(\theta)$ as a single global "observation" contributing one deterministic score $s_\pi(\theta)$ and one deterministic curvature $\mathbf{H}_{\text{prior}} = -\nabla_\theta^2 \log \pi(\theta)$. This convention has the following consequences:

- At fixed θ, $s_\pi(\theta)$ is a constant under the data-sampling distribution; hence its sampling variance vanishes and its sampling covariance with any $s_t(\theta)$ is zero.
- The prior contributes additively to the curvature budget without introducing cross-interval terms.
- For a Gaussian prior, $\mathbf{H}_{\text{prior}}$ is constant in θ and diagonal in the transformed coordinates.
- When $\mathbf{H}_{\text{prior}}$ is positive definite (proper Gaussian prior) and $\mathbf{H}_{\text{lik}} \succeq 0$, the posterior curvature $\mathbf{H}_{\text{post}} = \mathbf{H}_{\text{lik}} + \mathbf{H}_{\text{prior}}$ developed in Section 2 is strictly positive definite, so the inverse square roots used in Sections 3–4 are well-defined without subspace projection (cf. Section 8 of the Likelihood supplement).

Posterior score mean and covariance under sampling. Combining (3) with the deterministic-prior-score convention,

$$\mathbb{E}[S_{\text{post}}(\theta)] = \mathbb{E}[S_{\text{lik}}(\theta)] + s_\pi(\theta), \tag{7}$$

$$\text{Cov}(S_{\text{post}}(\theta)) = \text{Cov}(S_{\text{lik}}(\theta)) = \mathbf{JT}_{\text{lik}}, \tag{8}$$

i.e. the prior shifts the mean of the posterior score but contributes nothing to its sampling covariance. The covariance contribution of the prior enters the diagnostic framework only through its curvature $\mathbf{H}_{\text{prior}}$, as defined in Section 3.

Important: $\mathbf{JT}_{\text{post}}$ is NOT $\text{Cov}(S_{\text{post}})$. Section 3 will define $\mathbf{JT}_{\text{post}} := \mathbf{JT}_{\text{lik}} + \mathbf{H}_{\text{prior}}$ by treating the prior as a virtual observation whose deterministic score contributes information $\mathbf{H}_{\text{prior}}$ to the curvature/fluctuation budget. By (8), this is *not* the literal covariance of S_{post}: $\mathbf{JT}_{\text{post}} \neq \text{Cov}(S_{\text{post}})$. This definitional choice is what makes the curvature/fluctuation identity $\mathbf{JT}_{\text{post}} = \mathbf{H}_{\text{post}}$ hold under correct likelihood specification regardless of prior strength.

Reduction to the likelihood case. When the prior becomes diffuse $\mathbf{H}_{\text{prior}} \to 0$ (formally an improper uniform prior, with $s_\pi \to 0$ as an automatic consequence of (5) at any fixed θ), equations (3)–(8) collapse to the corresponding likelihood definitions and $S_{\text{post}} \to S_{\text{lik}}$. When $\mathbf{H}_{\text{lik}}$ is full rank, the limit is regular and every Posterior IDM quantity defined in subsequent sections reduces smoothly to its Likelihood counterpart. When $\mathbf{H}_{\text{lik}}$ is singular, the limit recovers the subspace-projection regime of Section 8 of the Likelihood supplement; the regularizing role of a strictly positive $\mathbf{H}_{\text{prior}}$ is exactly what Posterior IDM exploits to avoid that projection.

2. Posterior Hessian Decomposition

The negative log-posterior decomposes as

$$-\log p(\theta \mid D) = -\ell(\theta) - \log \pi(\theta) + \text{const}, \tag{9}$$

so its expected curvature splits additively,

$$\mathbf{H}_{\text{post}}(\theta) = \mathbf{H}_{\text{lik}}(\theta) + \mathbf{H}_{\text{prior}}(\theta). \tag{10}$$

The likelihood term $\mathbf{H}_{\text{lik}}$ is the Gaussian Fisher Information of Section 2 of the Likelihood supplement:

$$\mathbf{H}_{\text{lik}}(\theta) = \sum_{t=1}^{T} \mathbb{E}\left[\mathbf{I}_t^{\text{G}}(\theta)\right] = -\mathbb{E}\left[\nabla_\theta^2 \ell(\theta)\right], \tag{11}$$

where the second equality is the Fisher identity, valid under correct likelihood specification and standard regularity (interchange of differentiation and expectation). The diagnostic role of $\mathbf{H}_{\text{lik}}$ versus $\mathbf{JT}_{\text{lik}}$ (Section 3 below) is to quantify departures from this identity.

The prior term is the precision matrix of the Gaussian prior in the transformed coordinates,

$$\mathbf{H}_{\text{prior}}(\theta) = -\nabla_\theta^2 \log \pi(\theta) = \Sigma_{\text{prior}}^{-1}, \tag{12}$$

which is constant in θ. When the prior factorizes across the transformed coordinates, $\mathbf{H}_{\text{prior}}$ is diagonal with entries $1/\sigma_{\text{prior},i}^2$.

The matrix $\mathbf{H}_{\text{post}}$ defines the local quadratic geometry of the posterior at the MAP and reduces to $\mathbf{H}_{\text{lik}}$ in the flat-prior limit $\mathbf{H}_{\text{prior}} \to 0$ (Section 2.3).

2.1 Positive-Definiteness of $\mathbf{H}_{\text{post}}$

A proper Gaussian prior has $\mathbf{H}_{\text{prior}} \succ 0$. When the analytic Gaussian Fisher Information (11) is used, $\mathbf{H}_{\text{lik}} \succeq 0$ by construction (as a sum of expected outer-product terms). Then for any nonzero $v \in \mathbb{R}^d$,

$$v^\top \mathbf{H}_{\text{post}}\, v \;\geq\; v^\top \mathbf{H}_{\text{prior}}\, v \;\geq\; \lambda_{\min}(\mathbf{H}_{\text{prior}})\, \|v\|^2 \;>\; 0, \qquad \mathbf{H}_{\text{post}} \;\succeq\; \mathbf{H}_{\text{prior}} \;\succ\; 0, \tag{13}$$

so $\mathbf{H}_{\text{post}}$ is strictly positive definite even when $\mathbf{H}_{\text{lik}}$ is singular or rank-deficient. No subspace projection is required.

Indefinite estimator-based $\mathbf{H}_{\text{lik}}$. In practice $\mathbf{H}_{\text{lik}}$ is often approximated by a finite-difference Hessian whose estimator bias can produce negative eigenvalues along weakly identified directions. In that case the Loewner bound $\mathbf{H}_{\text{post}} \succeq \mathbf{H}_{\text{prior}}$ no longer follows; instead, positive definiteness of $\mathbf{H}_{\text{post}}$ requires the prior to dominate the negative spectrum of $\mathbf{H}_{\text{lik}}$.

Let the spectral decomposition of $\mathbf{H}_{\text{lik}}$ be

$$\mathbf{H}_{\text{lik}} = \mathbf{H}_{\text{lik}}^{+} - \mathbf{H}_{\text{lik}}^{-}, \qquad \mathbf{H}_{\text{lik}}^{+} \succeq \mathbf{0}, \qquad \mathbf{H}_{\text{lik}}^{-} \succeq \mathbf{0}, \tag{14}$$

where $\mathbf{H}_{\text{lik}}^{+}$ collects the contribution from non-negative eigenvalues and $\mathbf{H}_{\text{lik}}^{-}$ the (sign-flipped) contribution from negative eigenvalues, so that both parts are PSD by construction (Jordan decomposition into positive and negative parts of a symmetric matrix). The necessary-and-sufficient condition for $\mathbf{H}_{\text{post}} = \mathbf{H}_{\text{lik}} + \mathbf{H}_{\text{prior}}$ to be positive definite is

$$\mathbf{H}_{\text{prior}} - \mathbf{H}_{\text{lik}}^{-} \;\succ\; \mathbf{0}, \tag{15}$$

i.e., the prior precision must dominate the negative part of $\mathbf{H}_{\text{lik}}$ in the Loewner order. Equivalently, for every unit vector $v \in \mathbb{R}^d$,

$$v^\top \mathbf{H}_{\text{prior}}\, v \;>\; v^\top \mathbf{H}_{\text{lik}}^{-}\, v, \tag{16}$$

i.e., the prior precision in every direction v exceeds the negative part of $\mathbf{H}_{\text{lik}}$ in that same direction.

Operational sufficient condition. A simple sufficient condition that requires only two scalar eigenvalue computations is

$$\lambda_{\min}(\mathbf{H}_{\text{prior}}) \;>\; \lambda_{\max}(\mathbf{H}_{\text{lik}}^{-}) \;=\; \max\bigl(0,\, -\lambda_{\min}(\mathbf{H}_{\text{lik}})\bigr), \tag{17}$$

which is sharp (necessary and sufficient) when $\mathbf{H}_{\text{prior}}$ and $\mathbf{H}_{\text{lik}}$ are simultaneously diagonalizable, and otherwise a conservative bound. In the codebase, the prior is diagonal in the transformed coordinates while $\mathbf{H}_{\text{lik}}$ generally has off-diagonal entries, so (17) is conservative and (15) is the diagnostic of record. The sufficient bound (17) is convenient as a quick monitoring statistic, but a rejection by (17) must trigger the matrix check (15) (cheap: a single eigendecomposition of $\mathbf{H}_{\text{lik}}$ plus the eigenvalue test of $\mathbf{H}_{\text{prior}} - \mathbf{H}_{\text{lik}}^{-}$) before declaring $\mathbf{H}_{\text{post}}$ ill-defined.

2.2 Consequences for Posterior IDM

Equation (13) has two structural consequences that distinguish Posterior IDM from its Likelihood counterpart:

- All Posterior IDM quantities defined below ($\mathbf{C}_{\text{post}}$, $\mathbf{C}_{\text{sample,post}}$, $\mathbf{R}_{\text{post}}$, $\Sigma_{\text{DCC,post}}$, $\mathbf{b}_{\text{DIB,post}}$) are well-defined on the full parameter space whenever $\mathbf{H}_{\text{lik}} \succeq 0$ (analytic Gaussian Fisher case) or whenever (15) holds (finite-difference case). The subspace projection of Section 8 of the Likelihood supplement is unnecessary in either case.
- The whitening transform is computed via the spectral decomposition $\mathbf{H}_{\text{post}} = \mathbf{U}_{\text{post}}\, \boldsymbol{\Lambda}_{\text{post}}\, \mathbf{U}_{\text{post}}^\top$, yielding the symmetric inverse square root $\mathbf{H}_{\text{post}}^{-1/2} = \mathbf{U}_{\text{post}}\, \boldsymbol{\Lambda}_{\text{post}}^{-1/2}\, \mathbf{U}_{\text{post}}^\top$. Both per-eigendirection (eigenvalues, tr, det) and per-parameter ($[\mathbf{C}_{\text{post}}]_{ii}$) diagnostics are then basis-invariant and interpretable as in Section 5 of the Likelihood supplement; see Section 4.5 for the operational details.

In computations we never form $\mathbf{H}_{\text{lik}}$ and $\mathbf{H}_{\text{prior}}$ separately when a stable factor of $\mathbf{H}_{\text{post}}$ is required: we assemble the sum (10) first, then factor.

2.3 Limiting Regimes

Flat-prior limit. As $\mathbf{H}_{\text{prior}} \to 0$, (10) degenerates to $\mathbf{H}_{\text{post}} \to \mathbf{H}_{\text{lik}}$. Combined with $\mathbf{JT}_{\text{post}} \to \mathbf{JT}_{\text{lik}}$ and $\mathbf{JS}_{\text{post}} \to \mathbf{JS}_{\text{lik}}$ from Section 3, every Posterior IDM quantity reduces termwise to its Likelihood IDM counterpart. When $\mathbf{H}_{\text{lik}}$ is rank-deficient, this collapse is understood on the retained eigenspace of $\mathbf{H}_{\text{lik}}$ (the singular-Fisher machinery of Section 8 of the Likelihood supplement re-emerges as the appropriate generalization).

Informative-prior limit. In the opposite limit $\lambda_{\min}(\mathbf{H}_{\text{prior}}) \to \infty$, $\mathbf{H}_{\text{post}}$ is dominated by $\mathbf{H}_{\text{prior}}$, the posterior contracts to the prior, and the conditioning of $\mathbf{H}_{\text{post}}$ is set by the prior precision. All Posterior IDM quantities remain well-defined throughout this entire range.

3. Posterior Score Fluctuation Matrices

The diagnostic role played by $\mathbf{JT}_{\text{lik}}$ ($\equiv \mathbf{J}_{\text{total}}$) and $\mathbf{JS}_{\text{lik}}$ ($\equiv \mathbf{J}_{\text{sample}}$) for the likelihood (Section 3 of the Likelihood supplement) must be lifted to the posterior so that the comparison with $\mathbf{H}_{\text{post}}$ yields meaningful values.

Recall the posterior score (3). Taken over realizations of D at fixed θ, $s_\pi(\theta)$ is a constant; hence it has zero sampling variance and, as a consequence, zero sampling covariance with $S_{\text{lik}}(\theta)$. The raw posterior score covariance therefore satisfies

$$\text{Cov}(S_{\text{post}}(\theta)) = \text{Cov}(S_{\text{lik}}(\theta)) = \mathbf{JT}_{\text{lik}}, \tag{18}$$

i.e. it does *not* contain $\mathbf{H}_{\text{prior}}$, while the negative posterior Hessian $\mathbf{H}_{\text{post}} = \mathbf{H}_{\text{lik}} + \mathbf{H}_{\text{prior}}$ does. A direct comparison $\mathbf{H}_{\text{post}}^{-1/2}\mathbf{JT}_{\text{lik}}\mathbf{H}_{\text{post}}^{-1/2}$ would conflate prior shrinkage with genuine information distortion and would not equal $\mathbf{I}$ under correct likelihood specification.

3.1 Prior as a single deterministic observation

We adopt the convention of treating the prior as one additional observation, independent of the data, with precision $\mathbf{H}_{\text{prior}}$ and a score that is deterministic across data realizations (zero fluctuation in D at fixed θ). Under this convention, the prior contributes its precision to every information-style matrix but contributes no temporal cross-terms with the data, since it is fixed before D is observed. We accordingly define

$$\mathbf{JT}_{\text{post}} := \mathbf{JT}_{\text{lik}} + \mathbf{H}_{\text{prior}}, \tag{19}$$

$$\mathbf{JS}_{\text{post}} := \mathbf{JS}_{\text{lik}} + \mathbf{H}_{\text{prior}}. \tag{20}$$

$\mathbf{JT}_{\text{post}}$ is a defined object, not a covariance. By (18), $\mathbf{JT}_{\text{post}}$ is *not* equal to the literal covariance of $S_{\text{post}}(\theta)$ (which is $\mathbf{JT}_{\text{lik}}$). Definition (19) is the augmented-observation bookkeeping convention that places the prior on the same footing as the data: it contributes its curvature directly to the information budget rather than through sampling variance. This convention is what propagates the likelihood Bartlett identity to the posterior in (21) below.

Inheritance of the information identity. With (19), the posterior information identity inherits from the likelihood one. If $\mathbf{JT}_{\text{lik}} = \mathbf{H}_{\text{lik}}$ (correct likelihood specification, Section 4 of the Likelihood supplement), then

$$\mathbf{JT}_{\text{post}} = \mathbf{H}_{\text{lik}} + \mathbf{H}_{\text{prior}} = \mathbf{H}_{\text{post}}. \tag{21}$$

This identity holds by construction of $\mathbf{JT}_{\text{post}}$ (Eq. (19)), not as an independent Bartlett identity for the posterior score; it is the additive convention that propagates the likelihood identity. Crucially, the equality survives arbitrary rescaling of $\mathbf{H}_{\text{prior}}$: as the prior ranges from weakly to dominantly informative, $\mathbf{JT}_{\text{post}}$ and $\mathbf{H}_{\text{post}}$ change in lockstep, so the resulting $\mathbf{C}_{\text{post}} = \mathbf{I}$ in Section 4 holds regardless of prior strength.

3.2 Sample/total decomposition

There is no posterior per-interval score $s_{t,\text{post}}$; the prior is treated as a single extra observation, so the within-interval covariance sum in $\mathbf{JS}_{\text{post}}$ is over data intervals only. Since the prior pseudo-score is deterministic, $\text{Cov}(s_t, s_\pi) = 0$ for every data interval t and the cross-interval sum runs only over data indices. The decomposition of $\mathbf{JT}_{\text{lik}}$ into within-interval variance plus cross-interval covariance therefore lifts unchanged:

$$\mathbf{JT}_{\text{post}} = \mathbf{JS}_{\text{post}} + 2\sum_{i<j} \operatorname{Cov}(s_i, s_j), \qquad i, j \in \{1, \dots, T\}, \tag{22}$$

with s_t the per-interval data score of (4). The prior contributes $\mathbf{H}_{\text{prior}}$ identically to both $\mathbf{JT}_{\text{post}}$ and $\mathbf{JS}_{\text{post}}$, and therefore drops out of the difference $\mathbf{JT}_{\text{post}} - \mathbf{JS}_{\text{post}}$, which equals $2\sum_{i<j} \operatorname{Cov}(s_i, s_j)$ and involves only the data scores. Equation (22) isolates temporal dependence in the data score from the prior contribution, exactly as in the likelihood case.

3.3 Vanishing prior limit

As $\mathbf{H}_{\text{prior}} \to \mathbf{0}$,

$$\mathbf{JT}_{\text{post}} \to \mathbf{JT}_{\text{lik}}, \qquad \mathbf{JS}_{\text{post}} \to \mathbf{JS}_{\text{lik}}, \qquad \mathbf{H}_{\text{post}} \to \mathbf{H}_{\text{lik}}, \tag{23}$$

and all posterior distortion matrices defined in Section 4 reduce to their Likelihood IDM counterparts. When $\mathbf{H}_{\text{lik}}$ is full rank the limit is direct; when $\mathbf{H}_{\text{lik}}$ is rank-deficient, the limit must be interpreted on its retained eigenspace (Section 8 of the Likelihood supplement). For strictly positive $\mathbf{H}_{\text{prior}}$, by contrast, the Posterior IDM is well-defined globally without any retained-subspace construction; this is the structural advantage exploited in Section 4.

4. Posterior Information Distortion (PID)

The central Bayesian diagnostic is the total Posterior Information Distortion

$$\mathbf{C}_{\text{post}}(\theta) = \mathbf{H}_{\text{post}}^{-1/2} \mathbf{JT}_{\text{post}} \mathbf{H}_{\text{post}}^{-1/2}, \tag{24}$$

the direct counterpart of the Likelihood Information Distortion $\mathbf{C} = \mathbf{H}^{-1/2}\mathbf{J}_{\text{total}}\mathbf{H}^{-1/2}$ (Section 4 of the Likelihood supplement; recall the bridge (1) that identifies $\mathbf{H} \equiv \mathbf{H}_{\text{lik}}$ and $\mathbf{J}_{\text{total}} \equiv \mathbf{JT}_{\text{lik}}$), with the likelihood curvature and total score covariance replaced by their posterior counterparts

$$\mathbf{H}_{\text{post}} = \mathbf{H}_{\text{lik}} + \mathbf{H}_{\text{prior}}, \qquad \mathbf{JT}_{\text{post}} = \mathbf{JT}_{\text{lik}} + \mathbf{H}_{\text{prior}}. \tag{25}$$

Throughout, $\mathbf{H}_{\text{lik}}$, $\mathbf{H}_{\text{prior}}$, $\mathbf{JT}_{\text{lik}}$, and $\mathbf{JT}_{\text{post}}$ are symmetric, and the symmetric inverse square root $\mathbf{H}_{\text{post}}^{-1/2}$ is the unique positive definite root of $\mathbf{H}_{\text{post}}^{-1}$.

4.1 Consistency under correct likelihood specification

Under correct likelihood specification and standard regularity at the true parameter θ_0, the population information identity $\mathbf{JT}_{\text{lik}} = \mathbf{H}_{\text{lik}}$ holds; empirical plug-ins satisfy it only up to sampling fluctuation. Substituting into (25) gives (21), $\mathbf{JT}_{\text{post}} = \mathbf{H}_{\text{post}}$, and therefore

$$\mathbf{C}_{\text{post}} = \mathbf{H}_{\text{post}}^{-1/2} \mathbf{H}_{\text{post}} \mathbf{H}_{\text{post}}^{-1/2} = \mathbf{I}. \tag{26}$$

This holds for *any* proper Gaussian prior, informative or weak: the prior contribution $\mathbf{H}_{\text{prior}}$ enters $\mathbf{H}_{\text{post}}$ and $\mathbf{JT}_{\text{post}}$ symmetrically and cancels under the whitening transform.

The companion diagnostics $\mathbf{C}_{\text{sample,post}} = \mathbf{H}_{\text{post}}^{-1/2}\mathbf{JS}_{\text{post}}\mathbf{H}_{\text{post}}^{-1/2}$ and $\mathbf{R}_{\text{post}} = \mathbf{JS}_{\text{post}}^{-1/2}\mathbf{JT}_{\text{post}}\mathbf{JS}_{\text{post}}^{-1/2}$ play the same role for within-interval mismatch and temporal correlation as $\mathbf{C}_{\text{sample}}$ and $\mathbf{R}$ do in the Likelihood supplement. Under correct specification with no temporal correlation, $\mathbf{C}_{\text{post}} = \mathbf{C}_{\text{sample,post}} = \mathbf{R}_{\text{post}} = \mathbf{I}$; under correct specification with temporal correlation, $\mathbf{C}_{\text{sample,post}} = \mathbf{I}$ and $\mathbf{R}_{\text{post}} \neq \mathbf{I}$, with $\mathbf{C}_{\text{post}}$ tracking $\mathbf{R}_{\text{post}}$ (the formal development is deferred to the sample/correlation decomposition section below).

4.2 Deviation as a pure likelihood diagnostic

If the likelihood is misspecified, $\mathbf{JT}_{\text{lik}} \neq \mathbf{H}_{\text{lik}}$ and

$$\mathbf{JT}_{\text{post}} - \mathbf{H}_{\text{post}} = \mathbf{JT}_{\text{lik}} - \mathbf{H}_{\text{lik}}, \tag{27}$$

so the prior contribution drops out of the unwhitened residual. Equivalently,

$$\mathbf{C}_{\text{post}} - \mathbf{I} = \mathbf{H}_{\text{post}}^{-1/2}\big(\mathbf{JT}_{\text{lik}} - \mathbf{H}_{\text{lik}}\big)\mathbf{H}_{\text{post}}^{-1/2}. \tag{28}$$

The numerator $\mathbf{JT}_{\text{lik}} - \mathbf{H}_{\text{lik}}$ depends only on likelihood quantities, so any nonzero deviation $\mathbf{C}_{\text{post}} \neq \mathbf{I}$ originates purely on the likelihood side: the prior cannot create distortion of this kind. The whitening matrix $\mathbf{H}_{\text{post}}^{-1/2}$ does, however, rescale the magnitude of the deviation: a strong prior shrinks $\mathbf{C}_{\text{post}} - \mathbf{I}$ toward zero (in the limit $\lambda_{\min}(\mathbf{H}_{\text{prior}}) \to \infty$, $\mathbf{C}_{\text{post}} \to \mathbf{I}$), while a weak prior recovers the Likelihood IDM magnitude $\mathbf{C} - \mathbf{I} = \mathbf{H}_{\text{lik}}^{-1/2}(\mathbf{JT}_{\text{lik}} - \mathbf{H}_{\text{lik}})\mathbf{H}_{\text{lik}}^{-1/2}$ (Section 4.3). Thus $\mathbf{C}_{\text{post}}$ diagnoses purely likelihood-side miscalibration, with a magnitude calibrated by total posterior curvature.

4.3 Flat-prior limit

As $\mathbf{H}_{\text{prior}} \to \mathbf{0}$,

$$\mathbf{H}_{\text{post}} \to \mathbf{H}_{\text{lik}}, \qquad \mathbf{JT}_{\text{post}} \to \mathbf{JT}_{\text{lik}}, \qquad \mathbf{C}_{\text{post}} \to \mathbf{C}, \tag{29}$$

recovering the Likelihood IDM of Section 4 of the Likelihood supplement. When $\mathbf{H}_{\text{lik}}$ is full rank this limit is literal; when $\mathbf{H}_{\text{lik}}$ is rank-deficient it is understood on its retained eigenspace (Section 8 of the Likelihood supplement), and the Posterior IDM with a vanishing-but-proper prior provides a regularized version of that subspace construction. The Posterior IDM is therefore a strict generalization that reduces to its frequentist analogue in the noninformative limit.

4.4 Well-posedness without subspace projection

For any proper Gaussian prior, $\mathbf{H}_{\text{prior}} \succ 0$ (diagonal in transformed coordinates). Combined with $\mathbf{H}_{\text{lik}} \succeq 0$ for the analytic Gaussian Fisher Information, Eq. (13) gives

$$\mathbf{H}_{\text{post}} \succeq \mathbf{H}_{\text{prior}} \succ \mathbf{0}, \qquad \text{so} \qquad \mathbf{H}_{\text{post}}^{-1/2} \text{ in (24) is unconditionally defined.} \tag{30}$$

Hence $\mathbf{C}_{\text{post}}$ requires no retained-eigenspace construction. When $\mathbf{H}_{\text{lik}}$ is a finite-difference estimator that can develop negative eigenvalues, well-posedness instead requires the explicit dominance condition (15); for proper priors and bounded estimator error this is easily monitored. This contrasts with the Likelihood IDM, where singular $\mathbf{H}_{\text{lik}}$ forces the subspace treatment of Section 8 of the Likelihood supplement.

Numerical caveat. Mathematical well-posedness does not imply numerical robustness: directions along which both $\mathbf{H}_{\text{lik}} \approx 0$ and the prior precision is small produce $\mathbf{H}_{\text{post}}$ with large condition number, and $\mathbf{C}_{\text{post}}$ entries in those directions are sensitive to prior specification. We recommend reporting the condition number of $\mathbf{H}_{\text{post}}$ alongside any $\mathbf{C}_{\text{post}}$ diagnostic.

4.5 Computation

We never form $\mathbf{H}_{\text{post}}^{-1}$ explicitly. The symmetric inverse square root is obtained from the spectral decomposition

$$\mathbf{H}_{\text{post}} = \mathbf{U}_{\text{post}}\,\mathbf{\Lambda}_{\text{post}}\,\mathbf{U}_{\text{post}}^{\top}, \qquad \mathbf{H}_{\text{post}}^{-1/2} = \mathbf{U}_{\text{post}}\,\mathbf{\Lambda}_{\text{post}}^{-1/2}\,\mathbf{U}_{\text{post}}^{\top}, \tag{31}$$

where $\mathbf{U}_{\text{post}}$ is orthogonal and $\mathbf{\Lambda}_{\text{post}}$ is diagonal positive (modulo the dominance condition (15) in the finite-difference case). The whitening transform of (24) is then computed in retained-eigenspace coordinates: project $\mathbf{JT}_{\text{post}}$ onto $\mathbf{U}_{\text{post}}$, apply the diagonal scaling $\mathbf{\Lambda}_{\text{post}}^{-1/2}$ on both sides, and embed back through $\mathbf{U}_{\text{post}}$. This is numerically stable, produces a symmetric result, and degenerates gracefully when $\mathbf{H}_{\text{lik}}$ contributes a near-singular direction (the prior precision still fixes that direction; no subspace projection required as long as the dominance condition (15) holds).

Per-parameter vs. spectral diagnostics. Because the symmetric inverse square root (31) is used throughout, $\mathbf{C}_{\text{post}}$ as computed is the SPD matrix of (24), not a basis-dependent congruent variant. Both per-parameter diagonals $[\mathbf{C}_{\text{post}}]_{ii}$ and spectral invariants (λ_i, tr, det) are therefore well-defined and basis-invariant in the same sense as Section 5 of the Likelihood supplement:

- $[\mathbf{C}_{\text{post}}]_{ii} = 1$ indicates posterior consistency along parameter i; values > 1 (< 1) indicate under- (over-) estimated posterior uncertainty relative to the Bayesian curvature $\mathbf{H}_{\text{post}}$.
- $\lambda_i(\mathbf{C}_{\text{post}}) = 1$ indicates consistency along the i-th eigen-direction; deviations have the same inflation/deflation interpretation as in Section 5 of the Likelihood supplement.

5. Per-Parameter and Spectral Distortion

The Posterior Information Distortion $\mathbf{C}_{\text{post}}$ of Section 4 admits two complementary readouts: a per-parameter diagonal in the original parameter basis and a spectral profile in its eigenbasis. Both are well-defined and basis-invariant under the symmetric whitening of Section 4.5, in the same sense as Section 5 of the Likelihood supplement, but with the likelihood curvature and total score covariance replaced by their posterior counterparts of Eq. (25).

5.1 Per-Parameter Distortion

In the original parameter basis, the diagonal entries of $\mathbf{C}_{\text{post}}$ read as a per-parameter ratio of realized score fluctuation to Bayesian curvature-predicted variance:

$$[\mathbf{C}_{\text{post}}]_{ii} \sim \frac{\text{real fluctuation variance in direction } i}{\text{Gaussian-posterior predicted variance in direction } i}, \tag{32}$$

where the predicted variance is the direction-i component of $\mathbf{H}_{\text{post}}^{-1}$ and the fluctuation variance is the corresponding component of $\mathbf{JT}_{\text{post}}$, both measured in the whitened metric of (24). Strictly, $[\mathbf{C}_{\text{post}}]_{ii} = e_i^{\top}\mathbf{H}_{\text{post}}^{-1/2}\,\mathbf{JT}_{\text{post}}\,\mathbf{H}_{\text{post}}^{-1/2}e_i$, which coincides with the literal coordinate ratio $[\mathbf{JT}_{\text{post}}]_{ii}/[\mathbf{H}_{\text{post}}]_{ii}$ only when $\mathbf{H}_{\text{post}}$ is diagonal in the parameter basis; in general (32) is the corresponding quantity in the whitened metric. With this reading:

- $[\mathbf{C}_{\text{post}}]_{ii} = 1$: posterior consistency for parameter i.
- $[\mathbf{C}_{\text{post}}]_{ii} > 1$: under-estimation of posterior variance for parameter i (over-confident Gaussian posterior).
- $[\mathbf{C}_{\text{post}}]_{ii} < 1$: over-estimation of posterior variance for parameter i (under-confident Gaussian posterior).

Basis invariance of the diagonal. Because $\mathbf{C}_{\text{post}}$ is built from the *symmetric* inverse square root $\mathbf{H}_{\text{post}}^{-1/2}$ of Eq. (31), an orthogonal change of the original parameter basis $\mathbf{Q}$ acts congruently, $\mathbf{H}_{\text{post}} \mapsto \mathbf{Q}\,\mathbf{H}_{\text{post}}\,\mathbf{Q}^{\top}$ and $\mathbf{JT}_{\text{post}} \mapsto \mathbf{Q}\,\mathbf{JT}_{\text{post}}\,\mathbf{Q}^{\top}$, so $\mathbf{C}_{\text{post}} \mapsto \mathbf{Q}\,\mathbf{C}_{\text{post}}\,\mathbf{Q}^{\top}$. The diagonal $[\mathbf{C}_{\text{post}}]_{ii}$ in any chosen parameter basis and the spectrum of $\mathbf{C}_{\text{post}}$ are both invariant under this group action. A Cholesky whitening $\mathbf{H}_{\text{post}} = \mathbf{L}_{\text{post}}\mathbf{L}_{\text{post}}^{\top}$ would instead yield a basis-dependent congruent matrix whose diagonal mixes parameter directions through the lower-triangular structure of $\mathbf{L}_{\text{post}}$; the symmetric construction avoids this.

5.2 Spectral Distortion

Let $\lambda_i = \lambda_i(\mathbf{C}_{\text{post}})$ denote the eigenvalues of $\mathbf{C}_{\text{post}}$, ordered $\lambda_1 \geq \cdots \geq \lambda_d > 0$. By construction $\mathbf{C}_{\text{post}}$ is symmetric positive definite (because $\mathbf{H}_{\text{post}} \succ \mathbf{0}$ by Section 4.4 and $\mathbf{JT}_{\text{post}} = \mathbf{JT}_{\text{lik}} + \mathbf{H}_{\text{prior}} \succ \mathbf{0}$ as the sum of a PSD covariance and a positive prior precision; congruence preserves positive definiteness), so the spectrum is real and strictly positive. Each λ_i admits the same inflation/deflation reading as in Section 5 of the Likelihood supplement, applied along the corresponding eigen-direction of $\mathbf{C}_{\text{post}}$:

- $\lambda_i = 1$: perfect agreement between Bayesian curvature and posterior score fluctuation along eigen-direction i.
- $\lambda_i > 1$: variance inflation along eigen-direction i (the Gaussian posterior under-estimates uncertainty there).
- $\lambda_i < 1$: deflation along eigen-direction i (the Gaussian posterior over-estimates uncertainty there).

Global summaries. The determinant and trace of $\mathbf{C}_{\text{post}}$ collapse the spectrum to scalar diagnostics:

$$\det(\mathbf{C}_{\text{post}}) = \prod_{i=1}^{d} \lambda_i, \qquad \frac{1}{d}\,\mathrm{tr}(\mathbf{C}_{\text{post}}) = \frac{1}{d}\sum_{i=1}^{d} \lambda_i. \tag{33}$$

The determinant quantifies the overall posterior information-volume distortion: $\det(\mathbf{C}_{\text{post}}) = 1$ under the consistency identity (26), $\det(\mathbf{C}_{\text{post}}) > 1$ signals net volumetric under-estimation of posterior uncertainty, and $\det(\mathbf{C}_{\text{post}}) < 1$ signals net volumetric over-estimation. The average directional distortion $\mathrm{tr}(\mathbf{C}_{\text{post}})/d$ is the mean per-eigen-direction inflation, useful as a single lightweight summary when the spectrum is concentrated. The geometric-mean summary $\exp\!\big(\frac{1}{d}\log\det(\mathbf{C}_{\text{post}})\big)$ is the more robust alternative when the spectrum is heavy-tailed.

Correct-specification readout. Under the population information identity $\mathbf{JT}_{\text{lik}} = \mathbf{H}_{\text{lik}}$ at θ_0, Eq. (26) gives $\mathbf{C}_{\text{post}} = \mathbf{I}$, hence

$$[\mathbf{C}_{\text{post}}]_{ii} = 1\ \forall i, \qquad \lambda_i(\mathbf{C}_{\text{post}}) = 1\ \forall i, \qquad \det(\mathbf{C}_{\text{post}}) = 1, \qquad \mathrm{tr}(\mathbf{C}_{\text{post}})/d = 1, \tag{34}$$

regardless of prior strength: every per-parameter and spectral summary collapses to its consistency value.

5.3 Limiting Regimes

Flat-prior limit. As $\mathbf{H}_{\text{prior}} \to \mathbf{0}$, both whitening matrix and numerator in (24) converge to their Likelihood IDM counterparts (Section 4.3 above), and the per-parameter and spectral diagnostics reduce termwise:

$$[\mathbf{C}_{\text{post}}]_{ii} \to [\mathbf{C}]_{ii}, \qquad \lambda_i(\mathbf{C}_{\text{post}}) \to \lambda_i(\mathbf{C}), \qquad \det(\mathbf{C}_{\text{post}}) \to \det(\mathbf{C}), \qquad \operatorname{tr}(\mathbf{C}_{\text{post}})/d \to \operatorname{tr}(\mathbf{C})/d, \tag{35}$$

recovering Section 5 of the Likelihood supplement. Diagonal convergence is immediate from entry-wise convergence of $\mathbf{C}_{\text{post}}$; eigenvalue convergence follows from Weyl's inequality under operator-norm convergence of $\mathbf{C}_{\text{post}}$ to $\mathbf{C}$. When $\mathbf{H}_{\text{lik}}$ is rank-deficient the limit is on its retained eigenspace (Section 8 of the Likelihood supplement); a strictly positive $\mathbf{H}_{\text{prior}}$ is exactly what makes the per-parameter and spectral diagnostics globally well-defined throughout.

Strong-prior limit. Loewner comparisons between $\mathbf{H}_{\text{prior}}$ and $\mathbf{H}_{\text{lik}}$ alone do not control $\mathbf{C}_{\text{post}}$ per direction: the symmetric whitening $\mathbf{H}_{\text{post}}^{-1/2}(\cdot)\,\mathbf{H}_{\text{post}}^{-1/2}$ is a non-local operation, and the relevant magnitude is that of $\mathbf{H}_{\text{post}}$ itself relative to the residual $\mathbf{JT}_{\text{lik}} - \mathbf{H}_{\text{lik}}$. Quantitatively, the residual identity (28) gives

$$\|\mathbf{C}_{\text{post}} - \mathbf{I}\| \;\le\; \lambda_{\min}(\mathbf{H}_{\text{post}})^{-1}\,\big\|\mathbf{JT}_{\text{lik}} - \mathbf{H}_{\text{lik}}\big\|, \tag{36}$$

so as $\lambda_{\min}(\mathbf{H}_{\text{prior}}) \to \infty$ (with $\mathbf{JT}_{\text{lik}} - \mathbf{H}_{\text{lik}}$ bounded), $\mathbf{C}_{\text{post}} \to \mathbf{I}$ uniformly, hence $[\mathbf{C}_{\text{post}}]_{ii} \to 1$ and $\lambda_i(\mathbf{C}_{\text{post}}) \to 1$ for every i. The diagnostic reports values close to one in the strong-prior regime, where the likelihood-side mismatch is whitened away by the large $\mathbf{H}_{\text{post}}$, in agreement with Section 4.2.

Large-data (data-dominated) limit. Under N independent trajectories at fixed proper $\mathbf{H}_{\text{prior}}$, $\mathbf{H}_{\text{lik}} = O(N)$ and $\mathbf{JT}_{\text{lik}} = O(N)$ while $\mathbf{H}_{\text{prior}} = O(1)$. Hence $\mathbf{H}_{\text{post}}/N \to \mathbf{H}_{\text{lik}}/N$ and $\mathbf{JT}_{\text{post}}/N \to \mathbf{JT}_{\text{lik}}/N$ (with the prior contributions $O(1/N)$), so

$$[\mathbf{C}_{\text{post}}]_{ii} \to [\mathbf{C}]_{ii}, \qquad \lambda_i(\mathbf{C}_{\text{post}}) \to \lambda_i(\mathbf{C}), \qquad N \to \infty, \tag{37}$$

the data-dominated counterpart of (35). Together with the correct-specification readout (34) and the strong-prior limit (36), the per-parameter and spectral diagnostics behave coherently across all three boundary regimes: $\mathbf{C}_{\text{post}} = \mathbf{I}$ under correct specification (any prior), $\mathbf{C}_{\text{post}} \to \mathbf{I}$ under strong prior (any likelihood), $\mathbf{C}_{\text{post}} \to \mathbf{C}$ under data-dominated or flat-prior asymptotics. The full-rank caveats of Sections 4.3 (analytic Gaussian Fisher) and 4.4 (dominance condition (15) for finite-difference $\mathbf{H}_{\text{lik}}$) carry over without change.

6. Posterior Distortion-Corrected Covariance (DCC)

The Bayesian counterpart of the Likelihood DCC of Section 6 of the Likelihood supplement is obtained by repeating the first-order Taylor expansion argument at the maximum a posteriori (MAP) estimate $\hat{\theta}_{\text{MAP}}$ rather than at the MLE. As in Section 4, the prior contribution enters both the curvature ($\mathbf{H}_{\text{post}}$) and the information budget ($\mathbf{JT}_{\text{post}}$); the resulting sandwich form is the Posterior DCC, $\Sigma_{\text{DCC,post}}$ (formally defined in (40) below).

6.1 Taylor Expansion of the Posterior Score at the MAP

Expand the posterior score $S_{\text{post}}(\theta) = S_{\text{lik}}(\theta) + s_\pi(\theta)$ of (3) around the true parameter θ_0:

$$S_{\text{post}}(\hat{\theta}_{\text{MAP}}) \;\approx\; S_{\text{post}}(\theta_0) \;+\; \nabla_\theta S_{\text{post}}(\theta_0)\,(\hat{\theta}_{\text{MAP}} - \theta_0). \tag{38}$$

At the MAP, $S_{\text{post}}(\hat{\theta}_{\text{MAP}}) = 0$. Replacing the empirical likelihood Hessian by its expectation $-\mathbf{H}_{\text{lik}}(\theta_0)$ and noting that $\nabla_\theta s_\pi = -\mathbf{H}_{\text{prior}}$ is deterministic by the convention of Section 1 (so equals its own expectation), we obtain $\nabla_\theta S_{\text{post}}(\theta_0) \approx -\mathbf{H}_{\text{post}}(\theta_0)$ and

$$\hat{\theta}_{\text{MAP}} - \theta_0 \;\approx\; \mathbf{H}_{\text{post}}^{-1}\, S_{\text{post}}(\theta_0). \tag{39}$$

This is the Bayesian analogue of $(\hat{\theta} - \theta_0) \approx \mathbf{H}^{-1} S(\theta_0)$ in Section 6 of the Likelihood supplement, with $\mathbf{H}_{\text{post}}$ in place of $\mathbf{H}_{\text{lik}}$ and the prior score $s_\pi(\theta_0)$ folded additively into $S_{\text{post}}(\theta_0)$. The expansion assumes $\|\hat{\theta}_{\text{MAP}} - \theta_0\|$ small; in the strong-prior regime examined below this displacement contracts toward μ_{prior} rather than θ_0, so beyond that regime $\Sigma_{\text{DCC,post}}$ functions as a defined diagnostic (Section 6.2) rather than as the literal MAP-sampling covariance.

6.2 Sandwich Form: Definition of $\Sigma_{\text{DCC,post}}$

Taking sampling covariances on both sides of (39) and using $\text{Cov}(S_{\text{post}}) = \mathbf{JT}_{\text{lik}}$ from (8) (since s_π is deterministic) would give the literal MAP sampling covariance $\mathbf{H}_{\text{post}}^{-1}\,\mathbf{JT}_{\text{lik}}\,\mathbf{H}_{\text{post}}^{-1}$. Diagnostically, however, the prior must contribute to the information budget on the same footing as the data, exactly as in the construction of $\mathbf{JT}_{\text{post}}$ in Section 3.1. We therefore adopt the augmented-observation convention and define

$$\Sigma_{\text{DCC,post}} \;:=\; \mathbf{H}_{\text{post}}^{-1}\,\mathbf{JT}_{\text{post}}\,\mathbf{H}_{\text{post}}^{-1}, \tag{40}$$

the Posterior Distortion-Corrected Covariance. With $\mathbf{JT}_{\text{post}} = \mathbf{JT}_{\text{lik}} + \mathbf{H}_{\text{prior}}$ from (19), this is the direct counterpart of the Likelihood sandwich $\Sigma_{\text{DCC}} = \mathbf{H}^{-1}\,\mathbf{J}_{\text{total}}\,\mathbf{H}^{-1}$ of Section 6 of the Likelihood supplement, with both ends of the sandwich and the meat updated to their posterior counterparts.

Defined diagnostic versus literal sampling covariance. $\Sigma_{\text{DCC,post}}$ is a defined diagnostic, not the literal MAP sampling covariance. The literal covariance derived from (39) is $\mathbf{H}_{\text{post}}^{-1}\,\mathbf{JT}_{\text{lik}}\,\mathbf{H}_{\text{post}}^{-1}$ and differs from (40) by an additive prior-shrinkage term:

$$\Sigma_{\text{DCC,post}} \;-\; \mathbf{H}_{\text{post}}^{-1}\,\mathbf{JT}_{\text{lik}}\,\mathbf{H}_{\text{post}}^{-1} \;=\; \mathbf{H}_{\text{post}}^{-1}\,\mathbf{H}_{\text{prior}}\,\mathbf{H}_{\text{post}}^{-1}. \tag{41}$$

The two coincide under correct specification (where both reduce to $\mathbf{H}_{\text{post}}^{-1}$, Section 6.4) and in the flat-prior limit. The augmented-observation convention is what makes the consistency reduction (43) below hold for any prior strength.

6.3 Whitening Identity

Substituting the definition (24) of $\mathbf{C}_{\text{post}}$ into (40) yields the same algebraic identity as in the likelihood case,

$$\Sigma_{\text{DCC,post}} \;=\; \mathbf{H}_{\text{post}}^{-1/2}\,\mathbf{C}_{\text{post}}\,\mathbf{H}_{\text{post}}^{-1/2}, \tag{42}$$

the Posterior analogue of $\Sigma_{\text{DCC}} = \mathbf{H}^{-1/2}\,\mathbf{C}\,\mathbf{H}^{-1/2}$. The Posterior DCC is therefore the symmetric inverse whitening of the PID by $\mathbf{H}_{\text{post}}^{1/2}$: $\mathbf{C}_{\text{post}}$ encodes the directional distortion and $\mathbf{H}_{\text{post}}^{-1/2}$ rescales it back into parameter units.

6.4 Consistency Reduction: Laplace Posterior Covariance

Under the population information identity $\mathbf{JT}_{\text{lik}}(\theta_0) = \mathbf{H}_{\text{lik}}(\theta_0)$ at the true parameter (Section 4.1), $\mathbf{C}_{\text{post}} = \mathbf{I}$ by (26), and (42) collapses to

$$\Sigma_{\text{DCC,post}} \;=\; \mathbf{H}_{\text{post}}^{-1}, \tag{43}$$

the standard Laplace-approximation posterior covariance at the MAP. The Posterior DCC therefore reduces to the canonical Bayesian uncertainty quantification whenever the population likelihood-side information identity holds; departures from $\mathbf{H}_{\text{post}}^{-1}$ measure precisely the posterior-relevant share of likelihood miscalibration (Section 4.2).

6.5 Limiting Regimes

Flat-prior limit. As $\mathbf{H}_{\text{prior}} \to \mathbf{0}$, the substitutions (23) apply termwise to (40), giving

$$\Sigma_{\text{DCC,post}} \;\longrightarrow\; \mathbf{H}_{\text{lik}}^{-1}\,\mathbf{JT}_{\text{lik}}\,\mathbf{H}_{\text{lik}}^{-1} \;=\; \Sigma_{\text{DCC}}, \tag{44}$$

the Likelihood sandwich of Section 6 of the Likelihood supplement. When $\mathbf{H}_{\text{lik}}$ is full rank this limit is literal. When $\mathbf{H}_{\text{lik}}$ is rank-deficient, $\Sigma_{\text{DCC,post}}$ diverges along null directions of $\mathbf{H}_{\text{lik}}$ as $O(\lambda_{\min}(\mathbf{H}_{\text{prior}})^{-1})$; on the retained eigenspace of $\mathbf{H}_{\text{lik}}$ the limit recovers Σ_{DCC}, and the Posterior DCC with a strictly positive $\mathbf{H}_{\text{prior}}$ provides the regularized analogue of the subspace construction of Section 8 of the Likelihood supplement.

Strong-prior limit. In the opposite regime $\lambda_{\min}(\mathbf{H}_{\text{prior}}) \to \infty$, the residual bound (36) forces $\mathbf{C}_{\text{post}} \to \mathbf{I}$, so by (42) $\Sigma_{\text{DCC,post}} \to \mathbf{H}_{\text{post}}^{-1}$. A direct Neumann expansion of (40) sharpens this to

$$\Sigma_{\text{DCC,post}} \;=\; \mathbf{H}_{\text{prior}}^{-1} \;+\; \mathbf{H}_{\text{prior}}^{-1}\big(\mathbf{JT}_{\text{lik}} - 2\mathbf{H}_{\text{lik}}\big)\mathbf{H}_{\text{prior}}^{-1} \;+\; O\big(\lambda_{\min}(\mathbf{H}_{\text{prior}})^{-3}\big), \tag{45}$$

so the Posterior DCC contracts to $\mathbf{H}_{\text{prior}}^{-1}$ at rate $O(\lambda_{\min}(\mathbf{H}_{\text{prior}})^{-2})$ with a signed correction set by $\mathbf{JT}_{\text{lik}} - 2\mathbf{H}_{\text{lik}}$. Note that the deviation from the Laplace covariance is the prior-rescaled likelihood mismatch, $\Sigma_{\text{DCC,post}} - \mathbf{H}_{\text{post}}^{-1} = \mathbf{H}_{\text{prior}}^{-1}(\mathbf{JT}_{\text{lik}} - \mathbf{H}_{\text{lik}})\mathbf{H}_{\text{prior}}^{-1} + O(\lambda_{\min}(\mathbf{H}_{\text{prior}})^{-3})$, vanishing at the same absolute rate as the Laplace covariance itself; the diagnostic loses absolute sensitivity to likelihood miscalibration in this regime.

Large-data (data-dominated) limit. For N independent trajectories at fixed proper $\mathbf{H}_{\text{prior}}$, $\mathbf{H}_{\text{lik}}, \mathbf{JT}_{\text{lik}} = O(N)$ while $\mathbf{H}_{\text{prior}} = O(1)$, so

$$\Sigma_{\text{DCC,post}} \;=\; \mathbf{H}_{\text{lik}}^{-1}\,\mathbf{JT}_{\text{lik}}\,\mathbf{H}_{\text{lik}}^{-1} \;+\; O(N^{-2}) \;=\; \Sigma_{\text{DCC}} + O(N^{-2}), \tag{46}$$

i.e. $\Sigma_{\text{DCC,post}} \propto 1/N$, recovering the Likelihood DCC scaling of Section 7 of the Likelihood supplement (Bernstein–von Mises). The same conclusion transfers to the single-trajectory $T \to \infty$ limit under stationarity and ergodicity of the per-interval scores.

6.6 Well-Posedness

The Posterior DCC is well-defined whenever $\mathbf{H}_{\text{post}} \succ \mathbf{0}$. For analytic Gaussian Fisher $\mathbf{H}_{\text{lik}} \succeq 0$ and any proper Gaussian prior, this is automatic by (13): no subspace projection is needed, in contrast with Section 8 of the Likelihood supplement. For finite-difference $\mathbf{H}_{\text{lik}}$ with possibly negative eigenvalues, well-posedness reduces to the dominance condition (15) of Section 2.1. In both cases $\Sigma_{\text{DCC,post}}$ is computed via the spectral whitening of (31), applied symmetrically on both sides of $\mathbf{JT}_{\text{post}}$; explicit inversion of $\mathbf{H}_{\text{post}}$ is never required.

6.7 Role in the Bayesian Evidence Correction

The Posterior DCC defined here is the covariance matrix that enters the distortion correction to the Bayesian model evidence developed in the companion supplement `supplement_posterior_evidence.tex`. Specifically, $\Sigma_{\mathrm{DCC,post}}$ replaces $\mathbf{H}_{\mathrm{post}}^{-1}$ as the local posterior covariance entering the Laplace approximation determinant for $\log p(D)$, restoring calibration under likelihood miscalibration in the same way that Σ_{DCC} restores calibration of frequentist confidence regions in the Likelihood supplement. The explicit evidence-level formula and its consequences for Bayes factor comparison are deferred to that companion document; this section provides only the parameter-space object $\Sigma_{\mathrm{DCC,post}}$ on which the evidence-level correction is built.

7. Posterior Distortion-Induced Bias (DIB)

The bias counterpart of the posterior corrected covariance of Section 6 above quantifies the systematic displacement of the MAP estimator induced jointly by likelihood misspecification and prior pull. It is the direct Bayesian analogue of the Likelihood Distortion-Induced Bias of Section 7 of the Likelihood supplement, augmented by an additive prior-pull term that the likelihood-only construction cannot see.

7.1 Expected Posterior Score

Taking expectations in (3)–(5) under the data-sampling distribution at fixed θ gives

$$\mathbf{g}_{\mathrm{post}}(\theta) \;:=\; \mathbb{E}[S_{\mathrm{post}}(\theta)] \;=\; \mathbb{E}[S_{\mathrm{lik}}(\theta)] + s_\pi(\theta) \;=\; \mathbf{g}_{\mathrm{lik}}(\theta) - \mathbf{H}_{\mathrm{prior}}\big(\theta - \mu_{\mathrm{prior}}\big), \tag{47}$$

where $\mathbf{g}_{\mathrm{lik}}(\theta) := \mathbb{E}[S_{\mathrm{lik}}(\theta)]$ is the Likelihood DIB precursor (the $\mathbf{g}$ of Section 7 of the Likelihood supplement, under the bridge (1) extended to score means in the obvious way). The decomposition (47) is exact and separates the data-side score drift from the prior-pull term linearly.

Behavior at the true parameter. At the data-generating θ_0, the data score has zero expectation under correct likelihood specification and standard regularity,

$$\mathbf{g}_{\mathrm{lik}}(\theta_0) = \mathbf{0}, \tag{48}$$

so $\mathbf{g}_{\mathrm{post}}(\theta_0) = s_\pi(\theta_0) = -\mathbf{H}_{\mathrm{prior}}(\theta_0 - \mu_{\mathrm{prior}})$ is generally nonzero: a correctly specified likelihood does not eliminate the prior pull, which vanishes only when the prior mean coincides with the truth ($\mu_{\mathrm{prior}} = \theta_0$). Under likelihood misspecification, $\mathbf{g}_{\mathrm{lik}}(\theta_0) \neq \mathbf{0}$ and both contributions are present.

7.2 Posterior Distortion-Induced Bias

Apply the first-order MAP expansion of Section 6.1 above to the posterior score, using the realized expansion $\hat\theta_{\mathrm{MAP}} - \theta_0 \approx \mathbf{H}_{\mathrm{post}}^{-1} S_{\mathrm{post}}(\theta_0)$ of (39). Taking expectations under the data-sampling distribution at θ_0,

$$\mathbb{E}\Big[\hat\theta_{\mathrm{MAP}} - \theta_0\Big] \;\approx\; \mathbf{H}_{\mathrm{post}}^{-1}\, \mathbf{g}_{\mathrm{post}}(\theta_0). \tag{49}$$

We accordingly define the posterior distortion-induced bias

$$\mathbf{b}_{\mathrm{DIB,post}}(\theta) \;:=\; \mathbf{H}_{\mathrm{post}}^{-1}\, \mathbf{g}_{\mathrm{post}}(\theta) \;=\; \mathbf{H}_{\mathrm{post}}^{-1}\big(\mathbf{g}_{\mathrm{lik}}(\theta) + s_\pi(\theta)\big). \tag{50}$$

The two summands isolate the two physical mechanisms: $\mathbf{H}_{\mathrm{post}}^{-1}\,\mathbf{g}_{\mathrm{lik}}(\theta)$ is the likelihood-misspecification contribution (a posterior-rescaled version of the Likelihood DIB), and $\mathbf{H}_{\mathrm{post}}^{-1}\, s_\pi(\theta)$ is the prior-pull contribution, present even under correct likelihood specification whenever $\mu_{\text{prior}} \neq \theta_0$. Since θ_0 is unknown, the reported diagnostic substitutes the plug-in estimate $\hat{\theta}_{\text{MAP}}$ for θ in (50); under misspecification this corresponds to evaluating $\mathbf{g}_{\text{lik}}$ at the pseudo-true value $\theta_* = \arg\min \mathbb{E}[-\ell(\theta)]$ to leading order.

Well-posedness. For any proper Gaussian prior, $\mathbf{H}_{\text{post}}^{-1}$ is unconditionally defined (Section 2.1, Eq. (13)); for finite-difference $\mathbf{H}_{\text{lik}}$ it is defined whenever the dominance condition (15) holds. Thus $\mathbf{b}_{\text{DIB,post}}$ requires no subspace projection.

7.3 Sample-Size Scaling

For N independent trajectories of length T, the likelihood-side quantities scale with N while the prior is fixed:

$$\mathbf{H}_{\text{lik}} = O(N), \qquad \mathbf{JT}_{\text{lik}} = O(N), \qquad \mathbf{H}_{\text{prior}} = O(1), \qquad s_\pi(\theta) = O(1). \tag{51}$$

Under correct likelihood specification at θ_0, $\|S_{\text{lik}}(\theta_0)\| = O_p(\sqrt{N})$ while $\mathbb{E}[S_{\text{lik}}(\theta_0)] = \mathbf{0}$; under misspecification, persistent drift gives $\|\mathbf{g}_{\text{lik}}(\theta_0)\| = O(N)$. Substituting into (50) and using $\mathbf{H}_{\text{post}} = \mathbf{H}_{\text{lik}} + \mathbf{H}_{\text{prior}} = O(N)$ in directions of positive $\mathbf{H}_{\text{lik}}$ curvature,

$$\mathbf{b}_{\text{DIB,post}}(\theta_0) \;=\; \underbrace{\mathbf{H}_{\text{post}}^{-1}\,\mathbf{g}_{\text{lik}}(\theta_0)}_{O(1)\ \text{(misspec)}} \;+\; \underbrace{\mathbf{H}_{\text{post}}^{-1}\,s_\pi(\theta_0)}_{O(1/N)}. \tag{52}$$

That is, the prior-pull component is suppressed as $1/N$, while the likelihood-misspecification component approaches the Likelihood DIB asymptote $\mathbf{H}_{\text{lik}}^{-1}\,\mathbf{g}_{\text{lik}}$ with an $O(1/N)$ correction from $\mathbf{H}_{\text{prior}}$. When $\mathbf{H}_{\text{lik}}$ is rank-deficient, this asymptote is on its retained eigenspace; on the null directions of $\mathbf{H}_{\text{lik}}$ the prior precision continues to dominate $\mathbf{H}_{\text{post}}$ at every N and the corresponding components of $\mathbf{b}_{\text{DIB,post}}$ remain prior-controlled rather than approaching a Likelihood DIB limit. Comparison with the corrected variance, $\Sigma_{\text{DCC,post}} = O(1/N)$ ((46)), gives the familiar diagnostic: bias dominates variance as N grows, exactly as in Section 7 of the Likelihood supplement, with the prior-pull contribution additionally vanishing. The same scaling transfers to the single-trajectory $T \to \infty$ regime under stationarity and ergodicity of the per-interval scores.

7.4 Flat-Prior and Strong-Prior Limits

Flat-prior limit. As $\mathbf{H}_{\text{prior}} \to \mathbf{0}$, $s_\pi(\theta) \to \mathbf{0}$ from (5) and $\mathbf{H}_{\text{post}} \to \mathbf{H}_{\text{lik}}$ from (23), so

$$\mathbf{b}_{\text{DIB,post}} \;\longrightarrow\; \mathbf{H}_{\text{lik}}^{-1}\,\mathbf{g}_{\text{lik}} \;=\; \mathbf{b}_{\text{DIB}}, \tag{53}$$

the Likelihood DIB of Section 7 of the Likelihood supplement. When $\mathbf{H}_{\text{lik}}$ is rank-deficient and $\mathbf{g}_{\text{lik}}$ has nonzero projection onto its null space, $\mathbf{b}_{\text{DIB,post}}$ diverges as $O(\lambda_{\min}(\mathbf{H}_{\text{prior}})^{-1})$ along those directions; on the retained eigenspace of $\mathbf{H}_{\text{lik}}$ the limit recovers $\mathbf{b}_{\text{DIB}}$. For strictly positive $\mathbf{H}_{\text{prior}}$ the construction is global, as in Section 2.1.

Strong-prior limit. In the opposite regime $\lambda_{\min}(\mathbf{H}_{\text{prior}}) \to \infty$, the likelihood-misspecification term is dominated by the prior, $\mathbf{H}_{\text{post}}^{-1}\,\mathbf{g}_{\text{lik}}(\theta_0) = O(\lambda_{\min}(\mathbf{H}_{\text{prior}})^{-1}) \to \mathbf{0}$, while the prior-pull term saturates: using $s_\pi(\theta_0) = -\mathbf{H}_{\text{prior}}(\theta_0 - \mu_{\text{prior}})$ and $\mathbf{H}_{\text{post}}^{-1} \to \mathbf{H}_{\text{prior}}^{-1}$,

$$\mathbf{b}_{\text{DIB,post}}(\theta_0) \;\longrightarrow\; -(\theta_0 - \mu_{\text{prior}}), \tag{54}$$

i.e. the bias is the full shrinkage to the prior mean, independent of likelihood miscalibration. This is the boundary statement complementary to (53): weak priors recover the pure-likelihood diagnostic, strong priors collapse the MAP onto μ_{prior}.

7.5 Interpretation

The decomposition (50) makes the two physical sources of MAP displacement separately readable from the diagnostic, with $\theta = \hat{\theta}_{\mathrm{MAP}}$ used as the plug-in evaluation point (Section 7.2):

- $\mathbf{H}_{\mathrm{post}}^{-1}\, s_\pi(\hat{\theta}_{\mathrm{MAP}}) = -\mathbf{H}_{\mathrm{post}}^{-1}\mathbf{H}_{\mathrm{prior}}(\hat{\theta}_{\mathrm{MAP}} - \mu_{\mathrm{prior}})$ is the prior-pull contribution, a deterministic shrinkage toward μ_{prior} exactly computable from the prior hyperparameters and $\mathbf{H}_{\mathrm{post}}$.
- $\mathbf{H}_{\mathrm{post}}^{-1}\, \mathbf{g}_{\mathrm{lik}}(\hat{\theta}_{\mathrm{MAP}})$ is the likelihood-misspecification contribution and inherits the diagnostic role of $\mathbf{b}_{\mathrm{DIB}}$: it vanishes under correct specification (cf. (48)) and reports persistent score drift otherwise.

A nonzero $\mathbf{b}_{\mathrm{DIB,post}}$ whose prior-pull component dominates is a benign feature of Bayesian shrinkage, vanishing with N; a nonzero misspecification component is the structural concern that does not vanish with sample size and is reported alongside $\Sigma_{\mathrm{DCC,post}}$ as the bias-variance trade-off at the relevant data scale.

8. Comparison: Singular Likelihood Hessian vs. Proper Prior

Section 8 of the Likelihood supplement develops retained-eigenspace formulas for the Likelihood Information Distortion, DCC, DIB, and Correlation Distortion because $\mathbf{H}_{\mathrm{lik}}$ (the $\mathbf{H}$ of the Likelihood supplement, under the bridge (1)) can be singular or rank-deficient: only the subspace where $\mathbf{H}_{\mathrm{lik}}$ has eigenvalues above a numerical cutoff carries Gaussian Fisher information, and inverse powers of $\mathbf{H}_{\mathrm{lik}}$ must be restricted there. In the Posterior framework with a proper Gaussian prior, this construction is unnecessary: $\mathbf{H}_{\mathrm{post}} = \mathbf{H}_{\mathrm{lik}} + \mathbf{H}_{\mathrm{prior}}$ is positive definite on the full parameter space (Section 2.1), so every Posterior IDM quantity of Sections 4–7 is globally well-defined. The prior precision plays the role of regularizer that the retained-subspace projection plays in the Likelihood supplement, with the additional advantage that the prior eigenvalues provide an objective regularization scale while the Likelihood retained-eigenspace cutoff is a numerical heuristic.

8.1 Why no projection is required

For a proper Gaussian prior, $\mathbf{H}_{\mathrm{prior}} \succ \mathbf{0}$. Two cases arise depending on the sign profile of $\mathbf{H}_{\mathrm{lik}}$:

- *Analytic Gaussian Fisher Information.* $\mathbf{H}_{\mathrm{lik}} \succeq \mathbf{0}$ by construction, so $\mathbf{H}_{\mathrm{post}} \succeq \mathbf{H}_{\mathrm{prior}} \succ \mathbf{0}$ unconditionally (Eq. (13)). The symmetric inverse square root $\mathbf{H}_{\mathrm{post}}^{-1/2}$ used to whiten $\mathbf{JT}_{\mathrm{post}}$ and $\mathbf{JS}_{\mathrm{post}}$ is unconditionally defined on all of $\mathbb{R}^d$. Rank-deficient $\mathbf{H}_{\mathrm{lik}}$ (null and zero-eigenvalue directions) is handled trivially: any $\mathbf{H}_{\mathrm{prior}} \succ \mathbf{0}$ suffices.
- *Finite-difference* $\mathbf{H}_{\mathrm{lik}}$ *with indefinite spectrum.* The Jordan decomposition $\mathbf{H}_{\mathrm{lik}} = \mathbf{H}_{\mathrm{lik}}^{+} - \mathbf{H}_{\mathrm{lik}}^{-}$ of (14) isolates the negative part. Positive definiteness of $\mathbf{H}_{\mathrm{post}}$ is then equivalent to the dominance condition $\mathbf{H}_{\mathrm{prior}} - \mathbf{H}_{\mathrm{lik}}^{-} \succ \mathbf{0}$ of (15). This single matrix inequality is the bookkeeping condition that replaces the retained-subspace treatment of Section 8 of the Likelihood supplement: rather than discarding negatively-biased directions, the prior absorbs them.

In both cases, the symmetric inverse square root (31) of $\mathbf{H}_{\mathrm{post}}$ is computed by spectral decomposition on the full parameter space; the Posterior IDM, sample distortion, correlation distortion, DCC, and DIB of Sections 4–7 are then evaluated globally without subspace selection.

8.2 Connection between the regimes

The flat-prior and strong-prior regimes are continuously connected; the singular-likelihood case at vanishing prior is the only non-smooth boundary.

Weak prior, strong data: $\lambda_{\max}(\mathbf{H}_{\text{prior}}) \ll \lambda_{\min}(\mathbf{H}_{\text{lik}})$**, with** $\mathbf{H}_{\text{lik}}$ **full rank.** $\mathbf{H}_{\text{post}} \approx \mathbf{H}_{\text{lik}}$ and $\mathbf{JT}_{\text{post}} \approx \mathbf{JT}_{\text{lik}}$, so $\mathbf{C}_{\text{post}} \to \mathbf{C}$ entrywise. The Posterior IDM recovers the Likelihood IDM to leading order in $\mathbf{H}_{\text{prior}}$. Analogously $\mathbf{C}_{\text{sample,post}} \to \mathbf{C}_{\text{sample}}$, $\mathbf{R}_{\text{post}} \to \mathbf{R}$, $\Sigma_{\text{DCC,post}} \to \Sigma_{\text{DCC}}$, $\mathbf{b}_{\text{DIB,post}} \to \mathbf{b}_{\text{DIB}}$, as established in the flat-prior limits of Sections 4.3, 5.3, 6.5, and 7.4 above.

Strong prior, weak data: $\lambda_{\min}(\mathbf{H}_{\text{prior}}) \to \infty$**.** By the residual bound (36), $\mathbf{C}_{\text{post}} \to \mathbf{I}$ provided the likelihood-side mismatch $\mathbf{JT}_{\text{lik}} - \mathbf{H}_{\text{lik}}$ remains bounded (the regular-misspecification regime; if $\|\mathbf{JT}_{\text{lik}}\|$ itself blows up with the prior, the limit need not hold). Equivalently $\Sigma_{\text{DCC,post}} \to \mathbf{H}_{\text{prior}}^{-1}$ (Eq. (45)) and $\mathbf{b}_{\text{DIB,post}} \to -(\theta_0 - \mu_{\text{prior}})$ (Eq. (54)): the diagnostic correctly reports that inference along these directions is prior-dominated, with full shrinkage to the prior mean and absolute likelihood-miscalibration signal washed out.

Vanishing prior with singular $\mathbf{H}_{\text{lik}}$**.** If $\mathbf{H}_{\text{prior}} \to \mathbf{0}$ while $\mathbf{H}_{\text{lik}}$ has a non-trivial null space (or eigenvalues below the numerical cutoff), the limit is singular. On the retained eigenspace of $\mathbf{H}_{\text{lik}}$ (the columns of $\mathbf{U}_H$ whose eigenvalues exceed the cutoff, in the notation of Section 8 of the Likelihood supplement), the Posterior IDM converges to the Likelihood IDM. On the complementary null space, $\mathbf{H}_{\text{post}}^{-1/2}$ diverges as $O(\lambda_{\min}(\mathbf{H}_{\text{prior}})^{-1/2})$; the limit of the full matrix is undefined and the appropriate object is the retained-subspace Likelihood IDM $\mathbf{C}_H$ of Section 8 of the Likelihood supplement. The Posterior IDM with a strictly positive $\mathbf{H}_{\text{prior}}$ is the regularized version of this construction: it preserves the rank-deficient directions as "prior-dominated" rather than excising them.

8.3 Practical recommendation

The two diagnostics report complementary information and are simultaneously computable whenever both are well-defined:

- Use the *Likelihood IDM* (Sections 4–10 of the Likelihood supplement) when $\mathbf{H}_{\text{lik}}$ is well-conditioned and a prior-independent calibration is desired. The retained-subspace machinery of Section 8 of the Likelihood supplement is unavoidable when $\mathbf{H}_{\text{lik}}$ is singular or near-singular.
- Use the *Posterior IDM* (Sections 4–7 above) when $\mathbf{H}_{\text{lik}}$ is singular, near-singular, or indefinite (e.g. finite-difference Hessian along weakly identified directions), or when the diagnostic is to reflect the actual Bayesian posterior used for inference. The dominance condition (15) is the sole bookkeeping requirement; no eigenvalue cutoff is selected.
- In well-conditioned regimes both are reportable, and the prior-strength contrast $\mathbf{C} - \mathbf{C}_{\text{post}}$ quantifies the shrinkage effect of the prior on apparent calibration: by (28), $\mathbf{C}_{\text{post}} - \mathbf{I}$ is the residual $\mathbf{JT}_{\text{lik}} - \mathbf{H}_{\text{lik}}$ rescaled by $\mathbf{H}_{\text{post}}^{-1/2}$ rather than $\mathbf{H}_{\text{lik}}^{-1/2}$. Since $\mathbf{H}_{\text{post}} \succeq \mathbf{H}_{\text{lik}}$ in the Loewner order, the operator-norm bound $\|\mathbf{C}_{\text{post}} - \mathbf{I}\| \leq \|\mathbf{C} - \mathbf{I}\|$ holds (the per-direction comparison is not generally monotone because $\mathbf{JT}_{\text{lik}} - \mathbf{H}_{\text{lik}}$ is indefinite under misspecification).

The choice is a reporting decision, not a structural restriction: $\mathbf{C}$ diagnoses pure likelihood miscalibration in Gaussian Fisher geometry; $\mathbf{C}_{\text{post}}$ diagnoses the same miscalibration weighted by the actual posterior curvature available for inference.

9. Posterior Effective Sample Size and Correlation Distortion

The Likelihood Correlation Distortion $\mathbf{R}$ of Section 9 of the Likelihood supplement isolates the temporal-dependence content of the score from its within-interval magnitude by whitening $\mathbf{JT}_{\text{lik}}$ with $\mathbf{JS}_{\text{lik}}^{1/2}$ rather than with $\mathbf{H}_{\text{lik}}^{1/2}$. The Bayesian analogue uses the augmented fluctuation matrices $\mathbf{JT}_{\text{post}}$ and $\mathbf{JS}_{\text{post}}$ of Section 3.

9.1 Posterior Correlation Distortion

Define the *Posterior Correlation Distortion*

$$\mathbf{R}_{\text{post}}(\theta) \;:=\; \mathbf{JS}_{\text{post}}^{-1/2}\,\mathbf{JT}_{\text{post}}\,\mathbf{JS}_{\text{post}}^{-1/2}, \tag{55}$$

the direct counterpart of $\mathbf{R} = \mathbf{JS}_{\text{lik}}^{-1/2}\,\mathbf{JT}_{\text{lik}}\,\mathbf{JS}_{\text{lik}}^{-1/2}$ (Section 9 of the Likelihood supplement, under the bridge (1)). The symmetric inverse square root is computed by spectral decomposition of $\mathbf{JS}_{\text{post}}$ as in Section 4.5; $\mathbf{R}_{\text{post}}$ is SPD as a congruence transform of the SPD matrix $\mathbf{JT}_{\text{post}}$ by $\mathbf{JS}_{\text{post}}^{-1/2}$. Spectral invariants of $\mathbf{R}_{\text{post}}$ (eigenvalues, trace, determinant) and per-parameter diagonals $[\mathbf{R}_{\text{post}}]_{ii}$ in the fixed original parameter basis are well-defined and independent of the square-root algorithm choice (symmetric vs. Cholesky), in the same sense as Section 4.5.

Well-posedness. For any proper Gaussian prior, $\mathbf{H}_{\text{prior}} \succ \mathbf{0}$. Since $\mathbf{JS}_{\text{lik}} \succeq \mathbf{0}$ as a sum of within-interval score covariances (Section 3 of the Likelihood supplement),

$$\mathbf{JS}_{\text{post}} \;=\; \mathbf{JS}_{\text{lik}} + \mathbf{H}_{\text{prior}} \;\succeq\; \mathbf{H}_{\text{prior}} \;\succ\; \mathbf{0}, \tag{56}$$

so $\mathbf{JS}_{\text{post}}^{-1/2}$ is unconditionally defined, in contrast with $\mathbf{JS}_{\text{lik}}^{-1/2}$, which requires $\mathbf{JS}_{\text{lik}} \succ \mathbf{0}$ (and otherwise the retained-subspace treatment of Section 8 of the Likelihood supplement). For analytic Gaussian Fisher $\mathbf{JT}_{\text{lik}} = \mathrm{Cov}(S_{\text{lik}}) \succeq \mathbf{0}$, so $\mathbf{JT}_{\text{post}} \succeq \mathbf{H}_{\text{prior}} \succ \mathbf{0}$ and $\mathbf{R}_{\text{post}} \succ \mathbf{0}$ globally. For finite-difference estimators where $\mathbf{JT}_{\text{lik}}$ may be indefinite, the analogue of the dominance condition (15), $\mathbf{H}_{\text{prior}} - \mathbf{JT}_{\text{lik}}^{-} \succ \mathbf{0}$ (with $\mathbf{JT}_{\text{lik}}^{-}$ the negative part of the estimator), is required for global SPD-ness of $\mathbf{R}_{\text{post}}$; cf. Section 2.1.

9.2 Cancellation of the prior in the cross-interval residual

By the sample/total decomposition (22) of Section 3 (and its rearrangement),

$$\mathbf{JT}_{\text{post}} - \mathbf{JS}_{\text{post}} \;=\; \mathbf{JT}_{\text{lik}} - \mathbf{JS}_{\text{lik}} \;=\; 2\sum_{i<j}\mathrm{Cov}(s_i, s_j), \tag{57}$$

since the prior precision $\mathbf{H}_{\text{prior}}$ enters $\mathbf{JT}_{\text{post}}$ and $\mathbf{JS}_{\text{post}}$ identically and drops out of the difference (Section 3.2). The numerator of the unwhitened residual $\mathbf{JT}_{\text{post}} - \mathbf{JS}_{\text{post}}$ is therefore a pure data quantity, encoding the cross-interval covariance of the per-interval data scores s_t. The role of the prior in $\mathbf{R}_{\text{post}}$ is confined to the whitening transform $\mathbf{JS}_{\text{post}}^{-1/2}(\cdot)\,\mathbf{JS}_{\text{post}}^{-1/2}$, which rescales the residual but does not generate any cross-interval signal of its own.

9.3 Independent-score limit: $\mathbf{R}_{\text{post}} = \mathbf{I}$

If the data scores are temporally uncorrelated, $\mathrm{Cov}(s_i, s_j) = \mathbf{0}$ for all $i \neq j$, then $\mathbf{JT}_{\text{lik}} = \mathbf{JS}_{\text{lik}}$, whence $\mathbf{JT}_{\text{post}} = \mathbf{JS}_{\text{post}}$ and

$$\mathbf{R}_{\text{post}} \;=\; \mathbf{JS}_{\text{post}}^{-1/2}\,\mathbf{JS}_{\text{post}}\,\mathbf{JS}_{\text{post}}^{-1/2} \;=\; \mathbf{I}, \tag{58}$$

exactly as in the Likelihood case. The independent-score baseline is the same for both diagnostics: a proper prior does not by itself create temporal-correlation signal in $\mathbf{R}_{\text{post}}$.

9.4 Correlated-score regime: prior-softened temporal signal

When the cross-interval sum $\sum_{i<j} \operatorname{Cov}(s_i, s_j) \neq \mathbf{0}$ (the residual (57) is nonzero), $\mathbf{JT}_{\text{post}} \neq \mathbf{JS}_{\text{post}}$ and $\mathbf{R}_{\text{post}} \neq \mathbf{I}$. (Individual nonzero off-diagonal covariances can cancel in the sum and yield $\mathbf{R}_{\text{post}} = \mathbf{I}$ even with serial dependence; it is the residual sum that controls deviation from identity.) The deviation is governed by the same data-side residual that drives $\mathbf{R} - \mathbf{I}$ in the Likelihood case, but rescaled by the augmented whitening matrix. Subtracting the identity in (55) and using (57),

$$\mathbf{R}_{\text{post}} - \mathbf{I} \;=\; \mathbf{JS}_{\text{post}}^{-1/2} \left(\mathbf{JT}_{\text{lik}} - \mathbf{JS}_{\text{lik}}\right) \mathbf{JS}_{\text{post}}^{-1/2}. \tag{59}$$

The data residual $\mathbf{JT}_{\text{lik}} - \mathbf{JS}_{\text{lik}} = 2 \sum_{i<j} \operatorname{Cov}(s_i, s_j)$ is identical to the one in the Likelihood case; the prior contribution enters only through $\mathbf{JS}_{\text{post}}^{-1/2}$. In directions where $\mathbf{H}_{\text{prior}}$ is large relative to $\mathbf{JS}_{\text{lik}}$, the whitening is dominated by the prior precision and $\mathbf{R}_{\text{post}} - \mathbf{I}$ is shrunk toward zero; in directions where $\mathbf{H}_{\text{prior}}$ is small compared to $\mathbf{JS}_{\text{lik}}$, the whitening matches the Likelihood case and $\mathbf{R}_{\text{post}} - \mathbf{I} \approx \mathbf{R} - \mathbf{I}$. Informative-prior directions therefore have $\mathbf{R}_{\text{post}} - \mathbf{I}$ shrunk toward zero in operator norm relative to the corresponding $\mathbf{R} - \mathbf{I}$ from the Likelihood supplement.

9.5 Posterior Effective Sample Size

The variance-of-sum vs. sum-of-variances framework of Section 9 of the Likelihood supplement does *not* transfer literally: the posterior whitening matrix $\mathbf{JS}_{\text{post}} = \mathbf{JS}_{\text{lik}} + \mathbf{H}_{\text{prior}}$ contains the prior precision, so neither $\mathbf{JT}_{\text{post}}$ nor $\mathbf{JS}_{\text{post}}$ matches a variance-of-sum or sum-of-variances of any actual data sequence. The scalar summaries

$$\kappa_{\text{tr}}^{\text{post}} \;=\; \frac{1}{d}\operatorname{tr}(\mathbf{R}_{\text{post}}) \;=\; \frac{1}{d}\sum_{i=1}^{d} \lambda_i(\mathbf{R}_{\text{post}}), \tag{60}$$

$$\kappa_{\text{det}}^{\text{post}} \;=\; \exp\!\left(\frac{1}{d}\log\det(\mathbf{R}_{\text{post}})\right) \;=\; \left(\prod_{i=1}^{d} \lambda_i(\mathbf{R}_{\text{post}})\right)^{1/d} \tag{61}$$

are spectral summaries of $\mathbf{R}_{\text{post}}$ measuring, respectively, the average and geometric-mean per-eigen-direction inflation of the augmented within-interval budget $\mathbf{JS}_{\text{post}}$ by cross-interval DATA covariance $\mathbf{JT}_{\text{lik}} - \mathbf{JS}_{\text{lik}}$; the prior enters only through the $\mathbf{JS}_{\text{post}}$ whitening, not through the residual numerator. The corresponding posterior effective-sample-size summary is

$$T_{\text{eff}}^{\text{post}} \;=\; T/\kappa^{\text{post}}, \qquad \kappa^{\text{post}} \in \{\kappa_{\text{tr}}^{\text{post}}, \kappa_{\text{det}}^{\text{post}}\}, \tag{62}$$

chosen according to the spectral profile. $T_{\text{eff}}^{\text{post}}$ does not have the literal $\operatorname{Var}(L_X)/\sum_t \operatorname{Var}(X_t)$ interpretation of the Likelihood T_{eff}; it is a defined posterior diagnostic that measures the inflation of the per-direction posterior within-interval budget by cross-interval data covariance, not the total information available for inference.

Isotropic temporal dependence. If $\mathbf{R}_{\text{post}} \approx \kappa^{\text{post}}\mathbf{I}$, the data-side temporal dependence (as seen through the posterior whitening) acts isotropically and a single scalar effective sample size captures the diagnostic. Anisotropy in $\mathbf{R}_{\text{post}}$ indicates that some eigen-directions of the posterior fluctuation budget carry more cross-interval signal than others, and the scalar summaries (60)–(61) should be reported alongside the spectrum, exactly as in the Likelihood case.

9.6 Limiting Regimes

Throughout this subsection, T (or N) and $\mathbf{H}_{\text{prior}}$ are varied independently: each limit is taken with the other quantity held fixed unless stated otherwise.

Flat-prior limit. As $\mathbf{H}_{\text{prior}} \to \mathbf{0}$ at fixed $\mathbf{JS}_{\text{lik}}, \mathbf{JT}_{\text{lik}}$, $\mathbf{JS}_{\text{post}} \to \mathbf{JS}_{\text{lik}}$ and $\mathbf{JT}_{\text{post}} \to \mathbf{JT}_{\text{lik}}$ by (23), so

$$\mathbf{R}_{\text{post}} \longrightarrow \mathbf{R}, \qquad \kappa_{\text{tr}}^{\text{post}} \to \kappa_{\text{tr}}, \qquad \kappa_{\text{det}}^{\text{post}} \to \kappa_{\text{det}}, \tag{63}$$

recovering the Likelihood Correlation Distortion of Section 9 of the Likelihood supplement. When $\mathbf{JS}_{\text{lik}}$ is rank-deficient the limit is on its retained eigenspace (Section 8 of the Likelihood supplement uses its own $\mathbf{JS}_{\text{lik}}$-eigenspace construction $\mathbf{R}_{J_{\text{sample}}}$, distinct from the $\mathbf{H}$-eigenspace projection used for $\mathbf{C}$); on the null space of $\mathbf{JS}_{\text{lik}}$, $\mathbf{R}_{\text{post}}$ does not converge to $\mathbf{R}$ and stays prior-controlled at every $\mathbf{H}_{\text{prior}}$. A strictly positive $\mathbf{H}_{\text{prior}}$ regularizes both subspaces simultaneously and makes $\mathbf{R}_{\text{post}}$ globally well-defined.

Strong-prior limit. As $\lambda_{\min}(\mathbf{H}_{\text{prior}}) \to \infty$ at fixed T (and fixed $\mathbf{JT}_{\text{lik}}, \mathbf{JS}_{\text{lik}}$, hence fixed bounded residual $\mathbf{JT}_{\text{lik}} - \mathbf{JS}_{\text{lik}}$), the operator-norm bound

$$\|\mathbf{R}_{\text{post}} - \mathbf{I}\| \;\le\; \lambda_{\min}(\mathbf{JS}_{\text{post}})^{-1} \big\|\mathbf{JT}_{\text{lik}} - \mathbf{JS}_{\text{lik}}\big\| \;=\; O\big(\lambda_{\min}(\mathbf{H}_{\text{prior}})^{-1}\big) \tag{64}$$

forces $\mathbf{R}_{\text{post}} \to \mathbf{I}$, hence $\kappa_{\text{tr}}^{\text{post}}, \kappa_{\text{det}}^{\text{post}} \to 1$ and $T_{\text{eff}}^{\text{post}} \to T$. The diagnostic correctly reports that data-side temporal dependence has no posterior leverage in the prior-dominated regime, consistent with the strong-prior collapses of $\mathbf{C}_{\text{post}}$ (Section 5.3), $\Sigma_{\text{DCC,post}}$ (Section 6.5), and $\mathbf{b}_{\text{DIB,post}}$ (Section 7.4). $T_{\text{eff}}^{\text{post}} = T$ here reports prior dominance (the data-side temporal-dependence channel is whitened away) rather than data informativeness.

Large-data limit. For N independent trajectories at fixed proper $\mathbf{H}_{\text{prior}}$, $\mathbf{JT}_{\text{lik}}, \mathbf{JS}_{\text{lik}} = O(N)$ while $\mathbf{H}_{\text{prior}} = O(1)$, so $\mathbf{JS}_{\text{post}} = \mathbf{JS}_{\text{lik}}(1 + O(1/N))$ (and similarly for $\mathbf{JT}_{\text{post}}$), giving $\mathbf{R}_{\text{post}} = \mathbf{R} + O(1/N)$. The posterior correlation distortion thus reproduces the Likelihood correlation distortion in the data-dominated regime, again paralleling the corresponding $\mathbf{C}_{\text{post}}, \Sigma_{\text{DCC,post}}, \mathbf{b}_{\text{DIB,post}}$ behaviours. The convergence is on the retained eigenspace of $\mathbf{JS}_{\text{lik}}$; on persistent null directions of $\mathbf{JS}_{\text{lik}}$ independent of N, $\mathbf{R}_{\text{post}}$ remains prior-controlled at every N.

9.7 Relation to $\mathbf{C}_{\text{post}}$ and $\mathbf{C}_{\text{sample,post}}$

Under correct likelihood specification (Section 4.1, Eq. (26)), $\mathbf{JT}_{\text{lik}} = \mathbf{H}_{\text{lik}}$ at θ_0 and $\mathbf{C}_{\text{post}} = \mathbf{I}$. The companion within-interval identity $\mathbf{JS}_{\text{lik}} = \mathbf{H}_{\text{lik}}$ at the population level is the per-interval Bartlett identity (a per-observation property holding conditionally at each t, hence summing to the within-interval Fisher identity independently of cross-interval correlation); this is a strengthening of the total Fisher identity used in Section 4.1 and is local in time. Under the within-interval identity, $\mathbf{C}_{\text{sample,post}} = \mathbf{I}$, and:

- *Correct specification, no temporal correlation.* Both identities hold and $\mathbf{JT}_{\text{lik}} = \mathbf{JS}_{\text{lik}}$, so $\mathbf{R}_{\text{post}} = \mathbf{I} = \mathbf{C}_{\text{post}} = \mathbf{C}_{\text{sample,post}}$.
- *Correct specification, temporally correlated scores.* The within-interval identity gives $\mathbf{JS}_{\text{lik}} = \mathbf{H}_{\text{lik}}$, hence $\mathbf{JS}_{\text{post}} = \mathbf{H}_{\text{post}}$ and $\mathbf{C}_{\text{sample,post}} = \mathbf{I}$; under this identity, $\mathbf{R}_{\text{post}} = \mathbf{JS}_{\text{post}}^{-1/2}\,\mathbf{JT}_{\text{post}}\,\mathbf{JS}_{\text{post}}^{-1/2} = \mathbf{H}_{\text{post}}^{-1/2}\,\mathbf{JT}_{\text{post}}\,\mathbf{H}_{\text{post}}^{-1/2} = \mathbf{C}_{\text{post}}$, i.e. the two whitening operations coincide.

- *Within-interval misspecification.* $\mathbf{C}_{\text{sample,post}} \neq \mathbf{I}$ and the relation between $\mathbf{C}_{\text{post}}$ and $\mathbf{R}_{\text{post}}$ is mediated by $\mathbf{C}_{\text{sample,post}}$ via the sample/correlation decomposition (Section 10), which holds in the restricted sense developed there (commuting / Fisher-coordinate caveat).

The three diagnostics therefore separate three distinct posterior-side concerns: $\mathbf{C}_{\text{post}}$ measures total miscalibration, $\mathbf{C}_{\text{sample,post}}$ measures within-interval mismatch, and $\mathbf{R}_{\text{post}}$ isolates the temporal-dependence content of the data scores as seen through the posterior whitening.

10. Posterior Sample Distortion and Decomposition

The Posterior IDM $\mathbf{C}_{\text{post}}$ of Section 4 combines two structurally distinct sources of miscalibration: a within-interval mismatch between sample score variance and posterior curvature (the analogue of $\mathbf{C}_{\text{sample}}$ in Section 10 of the Likelihood supplement), and a cross-interval temporal-correlation effect (the analogue of $\mathbf{R}$, formalized as $\mathbf{R}_{\text{post}}$ in Section 9 above). This section promotes the within-interval object $\mathbf{C}_{\text{sample,post}}$ (introduced inline in Section 4.1 and listed among the Posterior IDM diagnostics in Section 2.2) to a labeled definition, states its consistency reduction, and establishes the sample/correlation decomposition

$$\mathbf{C}_{\text{post}} \;=\; \mathbf{M}\,\mathbf{R}_{\text{post}}\,\mathbf{M}^{\top} \;=\; \mathbf{C}^{1/2}_{\text{sample,post}}\,\widetilde{\mathbf{R}}_{\text{post}}\,\mathbf{C}^{1/2}_{\text{sample,post}}, \tag{65}$$

where $\mathbf{M}$ is the (non-orthogonal) basis change between the $\mathbf{JS}_{\text{post}}$-whitened and $\mathbf{H}_{\text{post}}$-whitened bases defined below, and $\widetilde{\mathbf{R}}_{\text{post}}$ is an orthogonal conjugate of $\mathbf{R}_{\text{post}}$ equal to $\mathbf{R}_{\text{post}}$ when $[\mathbf{H}_{\text{post}}, \mathbf{JS}_{\text{post}}] = \mathbf{0}$. This is the structural counterpart of $\mathbf{C} = \mathbf{C}^{1/2}_{\text{sample}}\,\mathbf{R}\,\mathbf{C}^{1/2}_{\text{sample}}$ of Section 10 of the Likelihood supplement (which carries the analogous commuting caveat), lifted to the augmented information budget of Section 3.

10.1 Posterior Sample Distortion

The Posterior Sample Distortion is the within-interval analogue of $\mathbf{C}_{\text{post}}$, obtained by replacing $\mathbf{JT}_{\text{post}}$ with $\mathbf{JS}_{\text{post}}$ in the whitening transform (24):

$$\mathbf{C}_{\text{sample,post}}(\theta) \;:=\; \mathbf{H}^{-1/2}_{\text{post}}\,\mathbf{JS}_{\text{post}}\,\mathbf{H}^{-1/2}_{\text{post}}. \tag{66}$$

By the augmented-observation convention of Section 3.1, $\mathbf{JS}_{\text{post}} = \mathbf{JS}_{\text{lik}} + \mathbf{H}_{\text{prior}}$ and $\mathbf{H}_{\text{post}} = \mathbf{H}_{\text{lik}} + \mathbf{H}_{\text{prior}}$ carry the prior precision identically; $\mathbf{C}_{\text{sample,post}}$ therefore quantifies the within-interval mismatch between data score variance and likelihood curvature. Well-posedness: $\mathbf{JS}_{\text{post}} \succeq \mathbf{H}_{\text{prior}} \succ \mathbf{0}$ for any proper Gaussian prior, so $\mathbf{JS}^{-1/2}_{\text{post}}$ (needed for both $\mathbf{R}_{\text{post}}$ and the map $\mathbf{M}$ below) is unconditionally defined, in parallel with $\mathbf{H}^{-1/2}_{\text{post}}$ of Section 4.4.

Consistency under correct within-interval specification. The per-interval Bartlett identity $E[s_t s_t^{\top} \mid y_{1:t-1}] = -E[\nabla s_t \mid y_{1:t-1}]$ holds conditionally for each t (a per-observation property of the conditional likelihood), and summing gives the within-interval Fisher identity $\mathbf{JS}_{\text{lik}} = \mathbf{H}_{\text{lik}}$ in expectation independently of cross-interval correlation. This strengthens the total Fisher identity used in Section 4.1 (which fails under serial dependence by Section 3.2) and is a genuine population statement: a derived per-interval consequence of correct conditional likelihood specification, not a quoted result from the Likelihood supplement. Substituting,

$$\mathbf{JS}_{\text{post}} \;=\; \mathbf{H}_{\text{lik}} + \mathbf{H}_{\text{prior}} \;=\; \mathbf{H}_{\text{post}}, \qquad \mathbf{C}_{\text{sample,post}} \;=\; \mathbf{I}. \tag{67}$$

This holds whether or not $\mathrm{Cov}(s_i, s_j) = 0$ for $i \neq j$: the temporal-correlation contribution lives entirely in the difference $\mathbf{JT}_{\text{post}} - \mathbf{JS}_{\text{post}} = 2\sum_{i<j}\mathrm{Cov}(s_i, s_j)$ from (22), hence in $\mathbf{R}_{\text{post}}$ rather than in $\mathbf{C}_{\text{sample,post}}$. The two diagnostics separate the two failure modes through the multiplicative decomposition (65) (additive only on the log-determinant scale, Section 10.5):

- $\mathbf{C}_{\text{sample,post}} \neq \mathbf{I}$ signals likelihood within-interval miscalibration ($\mathbf{JS}_{\text{lik}} \neq \mathbf{H}_{\text{lik}}$), the per-interval Fisher identity failing.
- $\mathbf{R}_{\text{post}} \neq \mathbf{I}$ signals cross-interval temporal correlation (the residual sum $\sum_{i<j}\mathrm{Cov}(s_i, s_j) \neq \mathbf{0}$), present under correct within-interval specification with serially dependent scores.

Correspondingly, $\mathbf{C}_{\text{post}} = \mathbf{I}$ only when both fail to fire, and otherwise $\mathbf{C}_{\text{post}}$ inherits its deviation from one or both sources jointly.

10.2 Decomposition Identity

Starting from the definitions (24) and (66), insert $\mathbf{JS}_{\text{post}}^{1/2}\,\mathbf{JS}_{\text{post}}^{-1/2} = \mathbf{I}$ on both sides of $\mathbf{JT}_{\text{post}}$:

$$\mathbf{C}_{\text{post}} \;=\; \mathbf{H}_{\text{post}}^{-1/2}\,\mathbf{JS}_{\text{post}}^{1/2}\left[\mathbf{JS}_{\text{post}}^{-1/2}\,\mathbf{JT}_{\text{post}}\,\mathbf{JS}_{\text{post}}^{-1/2}\right]\mathbf{JS}_{\text{post}}^{1/2}\,\mathbf{H}_{\text{post}}^{-1/2}. \tag{68}$$

The bracketed factor is $\mathbf{R}_{\text{post}}$ from Section 9. Setting

$$\mathbf{M} \;:=\; \mathbf{H}_{\text{post}}^{-1/2}\,\mathbf{JS}_{\text{post}}^{1/2}, \tag{69}$$

which is generally *not* symmetric (it is symmetric iff $\mathbf{H}_{\text{post}}$ and $\mathbf{JS}_{\text{post}}$ commute), Eq. (68) reads

$$\mathbf{C}_{\text{post}} \;=\; \mathbf{M}\,\mathbf{R}_{\text{post}}\,\mathbf{M}^\top, \qquad \mathbf{M}\,\mathbf{M}^\top \;=\; \mathbf{H}_{\text{post}}^{-1/2}\,\mathbf{JS}_{\text{post}}^{1/2}\,(\mathbf{JS}_{\text{post}}^{1/2})^\top\,(\mathbf{H}_{\text{post}}^{-1/2})^\top \;=\; \mathbf{H}_{\text{post}}^{-1/2}\,\mathbf{JS}_{\text{post}}\,\mathbf{H}_{\text{post}}^{-1/2} \;=\; \mathbf{C}_{\text{sample,post}}, \tag{70}$$

using symmetry of $\mathbf{JS}_{\text{post}}^{1/2}$ and $\mathbf{H}_{\text{post}}^{-1/2}$ in the third equality. So $\mathbf{C}_{\text{sample,post}} = \mathbf{M}\mathbf{M}^\top$: $\mathbf{M}$ is a non-symmetric square-root factor of $\mathbf{C}_{\text{sample,post}}$, related to the symmetric PSD root by the polar decomposition $\mathbf{M} = \mathbf{C}_{\text{sample,post}}^{1/2}\,\mathbf{Q}$ with $\mathbf{Q}$ orthogonal. The map $\mathbf{M}$ is the (generally non-orthogonal) basis change from the $\mathbf{JS}_{\text{post}}$-whitened basis (in which $\mathbf{R}_{\text{post}}$ is the natural object) to the $\mathbf{H}_{\text{post}}$-whitened basis (in which $\mathbf{C}_{\text{post}}$ lives); conjugation by $\mathbf{M}$ (i.e. $B \mapsto \mathbf{M}B\mathbf{M}^\top$) sends $\mathbf{R}_{\text{post}}$ to $\mathbf{C}_{\text{post}}$.

Sandwich form and the polar-factor caveat. Substituting the polar decomposition into (70),

$$\mathbf{C}_{\text{post}} \;=\; \mathbf{C}_{\text{sample,post}}^{1/2}\,\widetilde{\mathbf{R}}_{\text{post}}\,\mathbf{C}_{\text{sample,post}}^{1/2}, \qquad \widetilde{\mathbf{R}}_{\text{post}} \;:=\; \mathbf{Q}\,\mathbf{R}_{\text{post}}\,\mathbf{Q}^\top, \tag{71}$$

i.e. the $\mathbf{C}_{\text{sample,post}}^{1/2}$-sandwich form holds with $\mathbf{R}_{\text{post}}$ replaced by its orthogonal conjugate $\widetilde{\mathbf{R}}_{\text{post}}$. The literal sandwich identity $\mathbf{C}_{\text{post}} = \mathbf{C}_{\text{sample,post}}^{1/2}\,\mathbf{R}_{\text{post}}\,\mathbf{C}_{\text{sample,post}}^{1/2}$ holds iff $\mathbf{Q} = \mathbf{I}$, equivalently iff $\mathbf{H}_{\text{post}}$ and $\mathbf{JS}_{\text{post}}$ commute (simultaneous diagonalizability), and in particular when $\mathbf{R}_{\text{post}} = \kappa\mathbf{I}$ is scalar (since $\mathbf{Q}\kappa\mathbf{I}\mathbf{Q}^\top = \kappa\mathbf{I}$). The parallel identity of Section 10 of the Likelihood supplement, stated there "in Fisher coordinates", carries the same commuting caveat. The matrices $\mathbf{C}_{\text{sample,post}}^{1/2}\,\mathbf{R}_{\text{post}}\,\mathbf{C}_{\text{sample,post}}^{1/2}$ and $\mathbf{C}_{\text{post}}$ are always congruent with the same inertia and the same determinant (Section 10.5), but they are not equal as matrices in general.

10.3 Effective Sample Size Correction

Isotropic temporal effect. If the temporal correlation acts isotropically in the $\mathbf{JS}_{\text{post}}$-whitened basis, $\mathbf{R}_{\text{post}} = \kappa\,\mathbf{I}$ for some scalar $\kappa > 0$, then orthogonal conjugation by $\mathbf{Q}$ leaves $\mathbf{R}_{\text{post}}$ invariant, so the polar-factor caveat of (71) drops out and

$$\mathbf{C}_{\text{post}} \;=\; \kappa\,\mathbf{C}_{\text{sample,post}}, \qquad \Sigma_{\text{DCC,post}} \;=\; \kappa\,\mathbf{H}_{\text{post}}^{-1/2}\,\mathbf{C}_{\text{sample,post}}\,\mathbf{H}_{\text{post}}^{-1/2}, \tag{72}$$

using (42). Under the additional within-interval Fisher identity ($\mathbf{C}_{\text{sample,post}} = \mathbf{I}$ by (67)), this collapses to

$$\Sigma_{\text{DCC,post}} \;=\; \kappa\,\mathbf{H}_{\text{post}}^{-1}, \tag{73}$$

so the Posterior DCC equals the Laplace covariance scaled by a single effective-sample-size factor κ. The familiar $N_{\text{eff}} = N/\kappa$ rescaling of frequentist standard errors lifts here to a scalar inflation of the entire posterior covariance matrix. The $\kappa\mathbf{I}$ ansatz applies in three limits established in Sections 4–8: (i) the strong-prior regime where $\mathbf{R}_{\text{post}} \to \mathbf{I}$ and $\mathbf{C}_{\text{sample,post}} \to \mathbf{I}$ jointly (Section 5.3), with $\kappa \to 1$; (ii) the within-interval Fisher identity (population correct specification, Section 10.1) combined with isotropic serial correlation; (iii) the flat-prior limit when the Likelihood κ is itself near-isotropic. The brackets on κ are: $\kappa \to \kappa_{\text{lik}}$ (the Likelihood ESS) as $\mathbf{H}_{\text{prior}} \to \mathbf{0}$, and $\kappa \to 1$ as $\lambda_{\min}(\mathbf{H}_{\text{prior}}) \to \infty$.

Anisotropic or sample-distorted regime. If $\mathbf{R}_{\text{post}}$ is anisotropic (temporal correlation direction-dependent) or $\mathbf{C}_{\text{sample,post}} \neq \mathbf{I}$ (within-interval likelihood mismatch), neither (72) nor (73) reduces to a scalar correction: the distortion is structurally direction-dependent and the full matrix (65), with attention to the polar-factor caveat (71), is required for inference. The two diagnostics serve as gates: report a scalar N_{eff} only when $\mathbf{R}_{\text{post}}$ is near-isotropic and $\mathbf{C}_{\text{sample,post}}$ is near $\mathbf{I}$ in operator norm; otherwise report $\Sigma_{\text{DCC,post}}$ directly.

10.4 Flat-Prior Limit

As $\mathbf{H}_{\text{prior}} \to \mathbf{0}$, the substitutions (23) apply to (66), (70), and (65), yielding termwise

$$\mathbf{C}_{\text{sample,post}} \to \mathbf{C}_{\text{sample}}, \qquad \mathbf{R}_{\text{post}} \to \mathbf{R}, \qquad \mathbf{C}_{\text{post}} \to \mathbf{C}, \tag{74}$$

recovering the decomposition of Section 10 of the Likelihood supplement (with the same polar-factor caveat). Rank caveats apply separately to the two whitening matrices: convergence $\mathbf{C}_{\text{sample,post}} \to \mathbf{C}_{\text{sample}}$ requires the $\mathbf{H}_{\text{lik}}$ retained-eigenspace construction (Section 8 of the Likelihood supplement); convergence $\mathbf{R}_{\text{post}} \to \mathbf{R}$ requires the distinct $\mathbf{JS}_{\text{lik}}$-eigenspace construction $\mathbf{R}_{J_{\text{sample}}}$ used there. The decomposition itself converges in the flat-prior limit on the joint retained eigenspace; off the population within-interval Fisher identity, this requires the retained eigenspaces of $\mathbf{H}_{\text{lik}}$ and $\mathbf{JS}_{\text{lik}}$ to coincide. A strictly positive $\mathbf{H}_{\text{prior}}$ regularizes both subspaces simultaneously and makes the decomposition (65) globally well-defined.

10.5 Strong-Prior Limit

As $\lambda_{\min}(\mathbf{H}_{\text{prior}}) \to \infty$ at fixed $\mathbf{H}_{\text{lik}}, \mathbf{JS}_{\text{lik}}, \mathbf{JT}_{\text{lik}}$: $\mathbf{H}_{\text{post}} \to \mathbf{H}_{\text{prior}}$, $\mathbf{JS}_{\text{post}} \to \mathbf{H}_{\text{prior}}$, and $\mathbf{JT}_{\text{post}} \to \mathbf{H}_{\text{prior}} + (\mathbf{JT}_{\text{lik}} - \mathbf{H}_{\text{lik}})$, so $\mathbf{H}_{\text{post}}$ and $\mathbf{JS}_{\text{post}}$ become asymptotically proportional (both approach $\mathbf{H}_{\text{prior}}$, in particular they commute in the limit) and

$$\mathbf{C}_{\text{sample,post}} \to \mathbf{I}, \qquad \mathbf{R}_{\text{post}} \to \mathbf{I}, \qquad \mathbf{Q} \to \mathbf{I}, \qquad \mathbf{C}_{\text{post}} \to \mathbf{I}, \tag{75}$$

each with rate $O(\lambda_{\min}(\mathbf{H}_{\text{prior}})^{-1})$ from the residual bound (36), in agreement with Section 5.3, Section 6.5, and Section 9.6. The decomposition (65) collapses trivially to $\mathbf{C}_{\text{post}} = \mathbf{I}$. Both $\mathbf{C}_{\text{sample,post}}$ and $\mathbf{R}_{\text{post}}$ lose absolute sensitivity to likelihood miscalibration and temporal correlation in this regime — a feature, not a failure mode, since inference along prior-dominated directions is governed by the prior and not by the data-side budget.

10.6 Evidence-Correction Determinant Split

Taking determinants directly from $\mathbf{C}_{\text{post}} = \mathbf{H}_{\text{post}}^{-1}\mathbf{JT}_{\text{post}}$ (in the similarity sense) and $\mathbf{C}_{\text{sample,post}} = \mathbf{H}_{\text{post}}^{-1}\mathbf{JS}_{\text{post}}$ (likewise),

$$\det(\mathbf{C}_{\text{post}}) \;=\; \frac{\det(\mathbf{JT}_{\text{post}})}{\det(\mathbf{H}_{\text{post}})} \;=\; \frac{\det(\mathbf{JS}_{\text{post}})}{\det(\mathbf{H}_{\text{post}})}\cdot\frac{\det(\mathbf{JT}_{\text{post}})}{\det(\mathbf{JS}_{\text{post}})} \;=\; \det(\mathbf{C}_{\text{sample,post}})\,\det(\mathbf{R}_{\text{post}}), \quad (76)$$

which is unconditional (it does not rely on the polar-factor caveat of (71): $\det(\mathbf{M}\mathbf{R}_{\text{post}}\mathbf{M}^\top) = \det(\mathbf{M})^2\det(\mathbf{R}_{\text{post}}) = \det(\mathbf{C}_{\text{sample,post}})\det(\mathbf{R}_{\text{post}})$). This is forward-relevant for the Bayesian evidence correction developed in `supplement_posterior_evidence.tex`: the IDM evidence correction $\frac{1}{2}\log\det(\mathbf{C}_{\text{post}})$ splits additively into a within-interval mismatch contribution $\frac{1}{2}\log\det(\mathbf{C}_{\text{sample,post}})$ and a temporal-correlation contribution $\frac{1}{2}\log\det(\mathbf{R}_{\text{post}})$. Both terms are computable separately from the same posterior IDM machinery; either can be reported as a standalone diagnostic of the corresponding failure mode, and the two are additive (not multiplicative) on the log-evidence scale. Under correct specification with no temporal correlation, both vanish; under correct within-interval specification with serial dependence only, the first vanishes and the second is the pure effective-sample-size correction; under within-interval misspecification only, the second vanishes and the first reports the per-interval Fisher mismatch.

Limiting behavior of the split. In the flat-prior limit, $\log\det(\mathbf{C}_{\text{sample,post}}) \to \log\det(\mathbf{C}_{\text{sample}})$ and $\log\det(\mathbf{R}_{\text{post}}) \to \log\det(\mathbf{R})$, recovering the Likelihood-supplement evidence split. In the strong-prior limit, both terms vanish at rate $O(\lambda_{\min}(\mathbf{H}_{\text{prior}})^{-2})$ by (75), in agreement with the strong-prior collapse of the absolute miscalibration signal documented in Section 6.5. The standalone-diagnostic recommendation in the previous paragraph is implicitly conditioned on the moderate-prior regime where each term is at its natural $O(1)$ scale.

11. Summary

This supplement lifts the Likelihood Information Distortion machinery of `supplement_information_distortion_m` to the Bayesian posterior by treating the prior as a single deterministic augmented observation (Section 1) and propagating its precision $\mathbf{H}_{\text{prior}}$ additively into both the curvature and the score-fluctuation budget (Sections 2–3). The resulting Posterior IDM family is globally well-defined for any proper Gaussian prior when $\mathbf{H}_{\text{lik}}$ is PSD (analytic Gaussian Fisher), and under the dominance condition (15) when $\mathbf{H}_{\text{lik}}$ is indefinite (finite-difference case), in both regimes without subspace projection (Section 8). It isolates pure likelihood-side miscalibration weighted by the actual posterior curvature available for inference.

11.1 Glossary

$\mathbf{H}_{\text{lik}} \to$ likelihood Gaussian Fisher curvature (Section 2);
bridges to $\mathbf{H}$ of the Likelihood supplement.

$\mathbf{H}_{\text{prior}} \to$ prior precision (Section 2);
deterministic, diagonal in transformed coordinates, absent from Likelihood IDM.

$\mathbf{H}_{\text{post}} \to \mathbf{H}_{\text{lik}} + \mathbf{H}_{\text{prior}},$ posterior Hessian (Section 2);
globally positive definite for proper prior or under the dominance condition (15).

$\mathbf{JT}_{\text{lik}} \to$ likelihood total score covariance (Section 3);
bridges to $\mathbf{J}_{\text{total}}$ of the Likelihood supplement.

$\mathbf{JS}_{\text{lik}} \to$ likelihood within-interval score covariance (Section 3);
bridges to $\mathbf{J}_{\text{sample}}$ of the Likelihood supplement.

$\mathbf{JT}_{\text{post}} \to \mathbf{JT}_{\text{lik}} + \mathbf{H}_{\text{prior}}$ (Section 3, augmented-observation convention);
not the literal $\text{Cov}(S_{\text{post}})$, which is $\mathbf{JT}_{\text{lik}}$.

$\mathbf{JS}_{\text{post}} \to \mathbf{JS}_{\text{lik}} + \mathbf{H}_{\text{prior}}$ (Section 3);
not the literal within-interval covariance of S_{post} (that is $\mathbf{JS}_{\text{lik}}$);
same prior augmentation as $\mathbf{JT}_{\text{post}}$, drops out of the difference $\mathbf{JT}_{\text{post}} - \mathbf{JS}_{\text{post}}$.

$\mathbf{C}_{\text{post}} \to \mathbf{H}_{\text{post}}^{-1/2}\, \mathbf{JT}_{\text{post}}\, \mathbf{H}_{\text{post}}^{-1/2},$ Posterior Information Distortion (Section 4);
$\mathbf{C}_{\text{post}} = \mathbf{I}$ under correct likelihood specification, any proper prior.

$\mathbf{C}_{\text{sample,post}} \to \mathbf{H}_{\text{post}}^{-1/2}\, \mathbf{JS}_{\text{post}}\, \mathbf{H}_{\text{post}}^{-1/2},$ Posterior Sample Distortion (Section 10);
within-interval mismatch only; by construction does not see cross-interval terms
(those are isolated in $\mathbf{R}_{\text{post}}$ via $\mathbf{JT}_{\text{post}} - \mathbf{JS}_{\text{post}}$).

$\mathbf{R}_{\text{post}} \to \mathbf{JS}_{\text{post}}^{-1/2}\, \mathbf{JT}_{\text{post}}\, \mathbf{JS}_{\text{post}}^{-1/2},$ Posterior Correlation Distortion (Section 9);
numerator of $\mathbf{R}_{\text{post}} - \mathbf{I}$ depends only on data scores; prior $\mathbf{H}_{\text{prior}}$
cancels from $\mathbf{JT}_{\text{post}} - \mathbf{JS}_{\text{post}}$ but rescales magnitude via $\mathbf{JS}_{\text{post}}^{-1/2}$.

$\Sigma_{\text{DCC,post}} \to \mathbf{H}_{\text{post}}^{-1}\, \mathbf{JT}_{\text{post}}\, \mathbf{H}_{\text{post}}^{-1},$ Posterior Distortion-Corrected Covariance (Section 6);
collapses to the Laplace covariance $\mathbf{H}_{\text{post}}^{-1}$ under correct specification.

$\mathbf{b}_{\text{DIB,post}}(\theta) \to \mathbf{H}_{\text{post}}^{-1}\big(\mathbf{g}_{\text{lik}}(\theta) + s_\pi(\theta)\big),$ Posterior Distortion-Induced Bias (Section 7);
decomposes into likelihood-misspecification and prior-pull components;
evaluated at θ_0 in the population derivation, at $\widehat{\theta}_{\text{MAP}}$ in the plug-in.

11.2 Where Posterior IDM differs from Likelihood IDM

Each Posterior IDM quantity reduces termwise to its Likelihood IDM counterpart in the flat-prior limit (Sections 4.3, 5.3, 6.5, 7.4) and in the data-dominated limit (Sections 5.3, 6.5, 7.3); when $\mathbf{H}_{\text{lik}}$ is rank-deficient the flat-prior reduction is on its retained eigenspace, and on null directions the Posterior IDM diverges as $O(\lambda_{\min}(\mathbf{H}_{\text{prior}})^{-1})$ as discussed in those sections. The structural differences from the Likelihood IDM are:

- *Global well-posedness.* For any proper Gaussian prior with PSD $\mathbf{H}_{\text{lik}}$, $\mathbf{H}_{\text{post}} \succ \mathbf{0}$ on all of $\mathbb{R}^d$ (Section 2.1), so the retained-eigenspace projection of Section 8 of the Likelihood supplement is unnecessary. For rank-deficient analytic $\mathbf{H}_{\text{lik}}$, the strictly positive $\mathbf{H}_{\text{prior}}$ absorbs the null directions with no ad hoc cutoff. For finite-difference $\mathbf{H}_{\text{lik}}$ (possibly indefinite), the dominance condition (15), $\mathbf{H}_{\text{prior}} - \mathbf{H}_{\text{lik}}^{-} \succ \mathbf{0}$, is the sole bookkeeping requirement that handles indefiniteness without the numerical eigenvalue cutoff of the Likelihood case (Sections 2.1, 8).

- *Prior cancels from the distortion residual.* By (28), $\mathbf{C}_{\text{post}} - \mathbf{I} = \mathbf{H}_{\text{post}}^{-1/2}(\mathbf{JT}_{\text{lik}} - \mathbf{H}_{\text{lik}})\mathbf{H}_{\text{post}}^{-1/2}$, so the numerator depends only on likelihood quantities. The same cancellation holds for the difference $\mathbf{JT}_{\text{post}} - \mathbf{JS}_{\text{post}} = \mathbf{JT}_{\text{lik}} - \mathbf{JS}_{\text{lik}}$ encoded by $\mathbf{R}_{\text{post}}$ (Sections 3.2 and 9). The prior contribution rescales the magnitude of the deviation (through $\mathbf{H}_{\text{post}}^{-1/2}$ or $\mathbf{JS}_{\text{post}}^{-1/2}$) but cannot create distortion of its own.

- *Posterior-curvature weighting.* A strong prior shrinks $\|\mathbf{C}_{\text{post}} - \mathbf{I}\|$ toward zero (Section 5.3) and contracts $\Sigma_{\text{DCC,post}}$ to $\mathbf{H}_{\text{prior}}^{-1}$ (Section 6.5); $\mathbf{b}_{\text{DIB,post}}$ evaluated at θ_0 saturates to the prior-mean shrinkage $-(\theta_0 - \mu_{\text{prior}})$ (Section 7.4), while its plug-in form at $\widehat{\theta}_{\text{MAP}}$ correspondingly contracts to zero as $\widehat{\theta}_{\text{MAP}} \to \mu_{\text{prior}}$. Both diagnostics lose absolute sensitivity to likelihood miscalibration in this regime (Sections 4.2, 6.5); this is a feature, not a failure mode, since inference along those directions is governed by the prior.

- *Bias decomposition.* $\mathbf{b}_{\text{DIB,post}}$ has a structural prior-pull term $\mathbf{H}_{\text{post}}^{-1} s_\pi$ (this supplement, Section 7.5) absent from $\mathbf{b}_{\text{DIB}}$; under the N-scaling of Section 7.3 this term is $O(1/N)$ along directions of positive $\mathbf{H}_{\text{lik}}$ curvature, while the likelihood-misspecification term inherits the $O(1)$ behaviour of the Likelihood DIB. Along null/weakly-identified directions of $\mathbf{H}_{\text{lik}}$, the prior-pull term remains $O(1)$ (Section 7.3).

11.3 Concluding remarks

Posterior IDM as natural Bayesian generalization. The Posterior IDM is the natural Bayesian counterpart of the Likelihood IDM: it is defined whenever $\mathbf{H}_{\text{post}} \succ \mathbf{0}$ (analytic Gaussian Fisher with proper prior, or finite-difference $\mathbf{H}_{\text{lik}}$ satisfying the dominance condition (15)), regardless of singularity or rank deficiency of $\mathbf{H}_{\text{lik}}$. Sections 4–7 and the sample/correlation decompositions of Sections 9–10 are stated globally on $\mathbb{R}^d$ with no subspace selection.

Consistency identity under any proper prior. Under correct likelihood specification at θ_0 and any proper Gaussian prior, the population information identity $\mathbf{JT}_{\text{lik}} = \mathbf{H}_{\text{lik}}$ combined with the augmented-observation convention (19) gives $\mathbf{JT}_{\text{post}} = \mathbf{H}_{\text{post}}$ by construction (not as an independent posterior-side Bartlett identity), hence $\mathbf{C}_{\text{post}} = \mathbf{I}$ and $\Sigma_{\text{DCC,post}} = \mathbf{H}_{\text{post}}^{-1}$ (the Laplace posterior covariance) *regardless of prior strength.* The prior contribution enters $\mathbf{H}_{\text{post}}$ and $\mathbf{JT}_{\text{post}}$ symmetrically and cancels under the whitening transform.

Diagnostic isolates pure likelihood-side miscalibration. Under likelihood misspecification, the deviation $\mathbf{C}_{\text{post}} - \mathbf{I} = \mathbf{H}_{\text{post}}^{-1/2}(\mathbf{JT}_{\text{lik}} - \mathbf{H}_{\text{lik}})\mathbf{H}_{\text{post}}^{-1/2}$ has a numerator that depends only on likelihood quantities, so the diagnostic isolates likelihood-side calibration failures weighted by the actual posterior curvature. The same cancellation applies to $\mathbf{JT}_{\text{post}} - \mathbf{JS}_{\text{post}}$, so the numerator of $\mathbf{R}_{\text{post}} - \mathbf{I}$ depends only on the data scores; the whitening by $\mathbf{JS}_{\text{post}}^{-1/2} = (\mathbf{JS}_{\text{lik}} + \mathbf{H}_{\text{prior}})^{-1/2}$ rescales the magnitude in proportion to the posterior within-interval budget but cannot create cross-interval distortion of its own.

Likelihood vs. Posterior IDM as a reporting choice. The choice between Likelihood IDM and Posterior IDM is a reporting decision, not a structural restriction. Reported in well-conditioned regimes, $\mathbf{C}$ diagnoses pure likelihood miscalibration in Gaussian Fisher geometry, while $\mathbf{C}_{\text{post}}$ diagnoses the same miscalibration weighted by the posterior curvature actually available for inference (Section 8.3). When $\mathbf{H}_{\text{lik}}$ is singular, near-singular, or indefinite (the latter subject to the dominance condition (15)), the Posterior IDM is the only globally well-defined report and replaces the retained-eigenspace machinery of Section 8 of the Likelihood supplement.

Companion documents. The definitions established in Sections 1–10 of this supplement underwrite three further analyses in this folder. The Posterior DCC of Section 6 enters the distortion correction to the Bayesian model evidence and the Bayes factor comparison developed in `supplement_posterior_evidence.tex`. The full Posterior IDM family enters the information-gain decomposition developed in `supplement_information_gain.tex`, separating data-driven from prior-driven contributions. The dominance condition (15), the well-posedness regimes of Section 8, and the bootstrap-level monitoring of $\mathbf{C}_{\text{post}}$, $\mathbf{R}_{\text{post}}$, and $\Sigma_{\text{DCC,post}}$ feed the safety-categorization policy of `supplement_safety_categorization.tex`. The Posterior IDM defined here is the parameter-space substrate on which all three companion documents build.